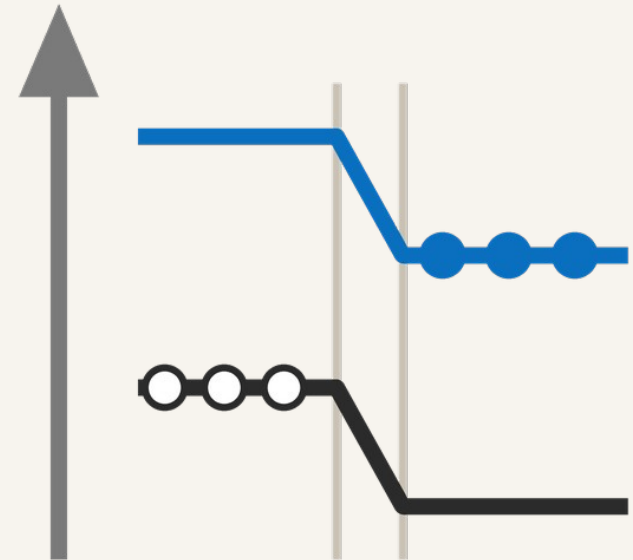


Semiconductors & p-n Junctions

Why some solids conduct, some don't,
and some can be talked into it.

Higher Physics



Starter

2009 Higher Physics

Scan the code and answer these:

Q8

Q11

Section A · charged particles, capacitors

Show your working, not just the letter.



**Scan for the paper
Marking scheme**

© Qualifications Scotland (SQA)

PART 1

Energy bands

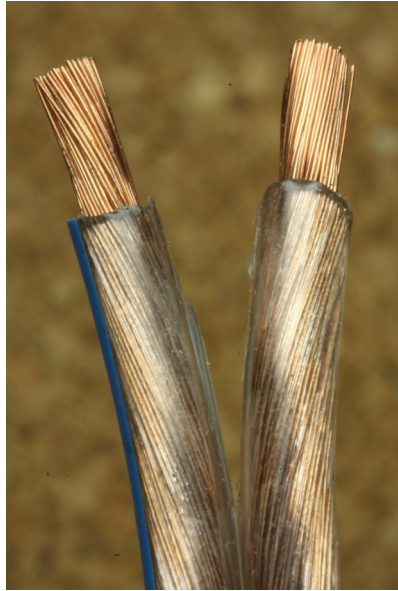
Why does copper conduct, glass not, and silicon only a little?

By the end of this part...

- 1 I can describe how energy bands form from the energy levels of single atoms.
- 2 I can name and describe the conduction band and the valence band.
- 3 I can use band structure to explain why a solid conducts or does not.

What makes a solid conduct?

Which of these conducts best, and which worst?



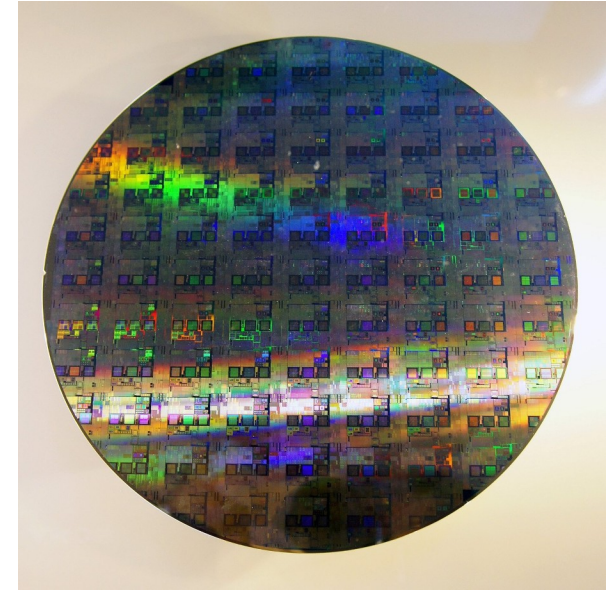
copper wire

Photo: [RobertKuhlmann](#)
CC BY-SA 3.0, Wikimedia Commons



glass insulators

Photo: [Maoileann](#)
CC BY-SA 4.0, Wikimedia Commons

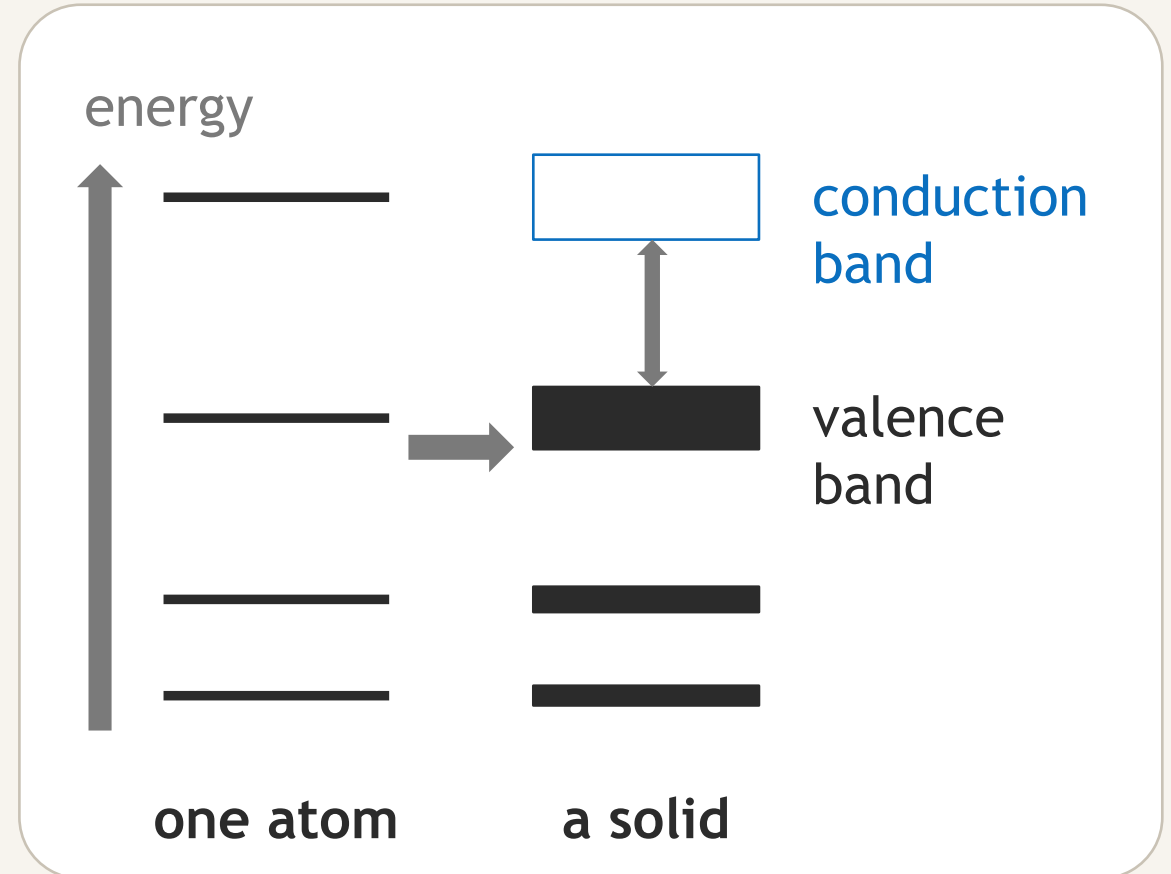


a silicon wafer

Photo: [PeelIden](#)
CC BY-SA 3.0, Wikimedia Commons

From energy levels to energy bands

- In a single atom, electrons sit in fixed energy levels.
- A solid is millions of atoms: the levels merge into bands.
- Between the bands are gaps no electron may have.

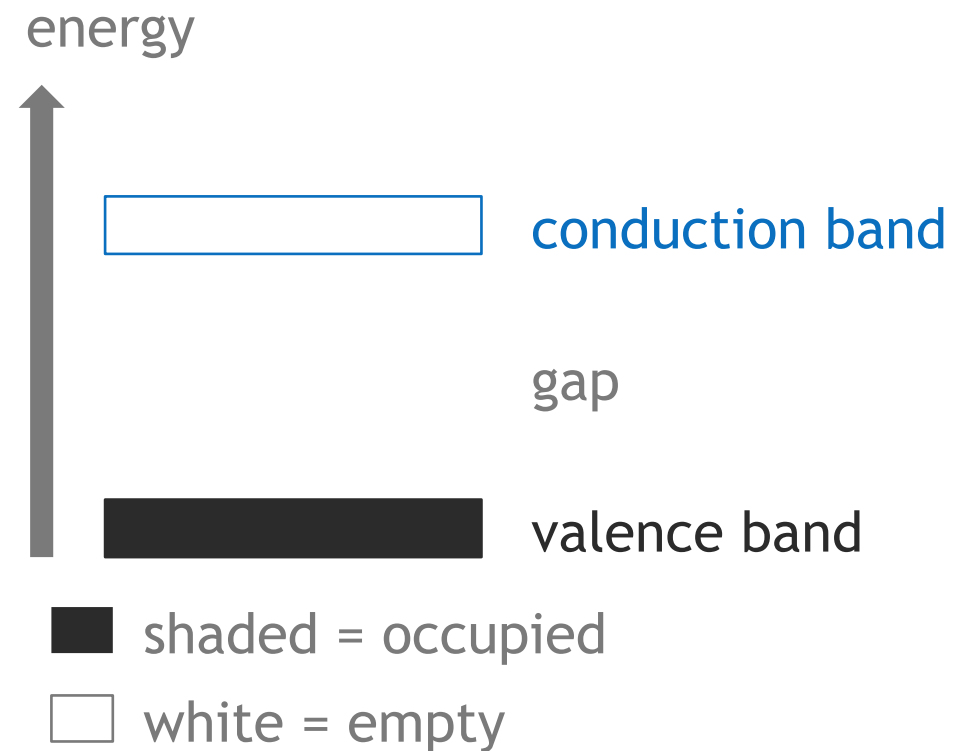




The conduction band and the valence band

The valence band holds the electrons that bond the solid together.

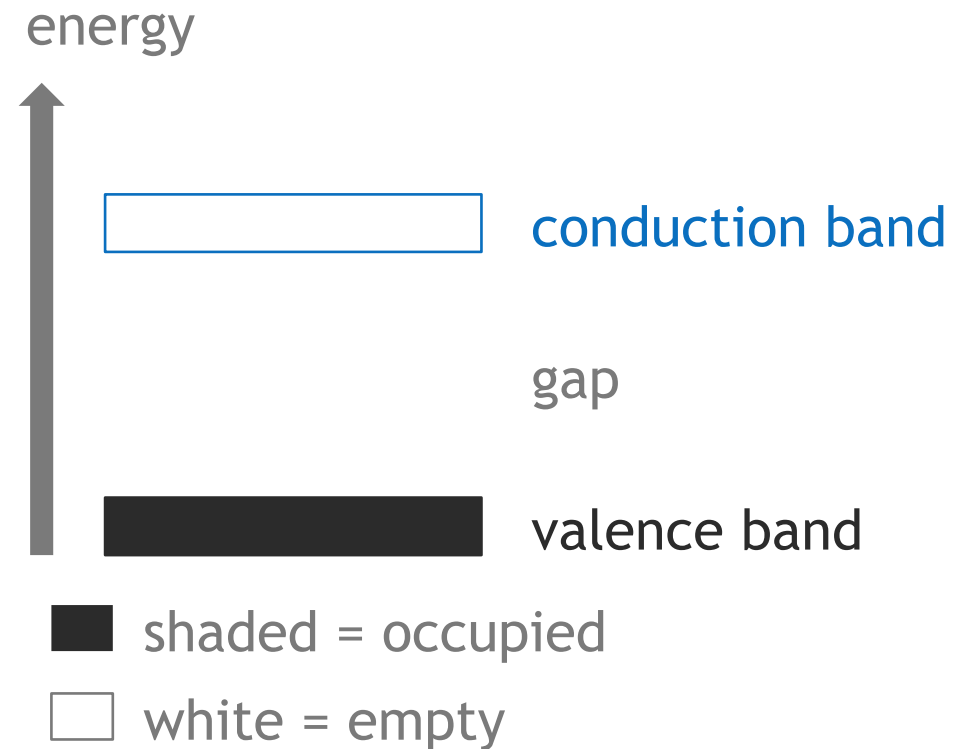
The conduction band is the band an electron has to reach before it can move freely and carry a current.





The gap between the bands

Between them is a gap – a range of energies no electron is allowed to have.

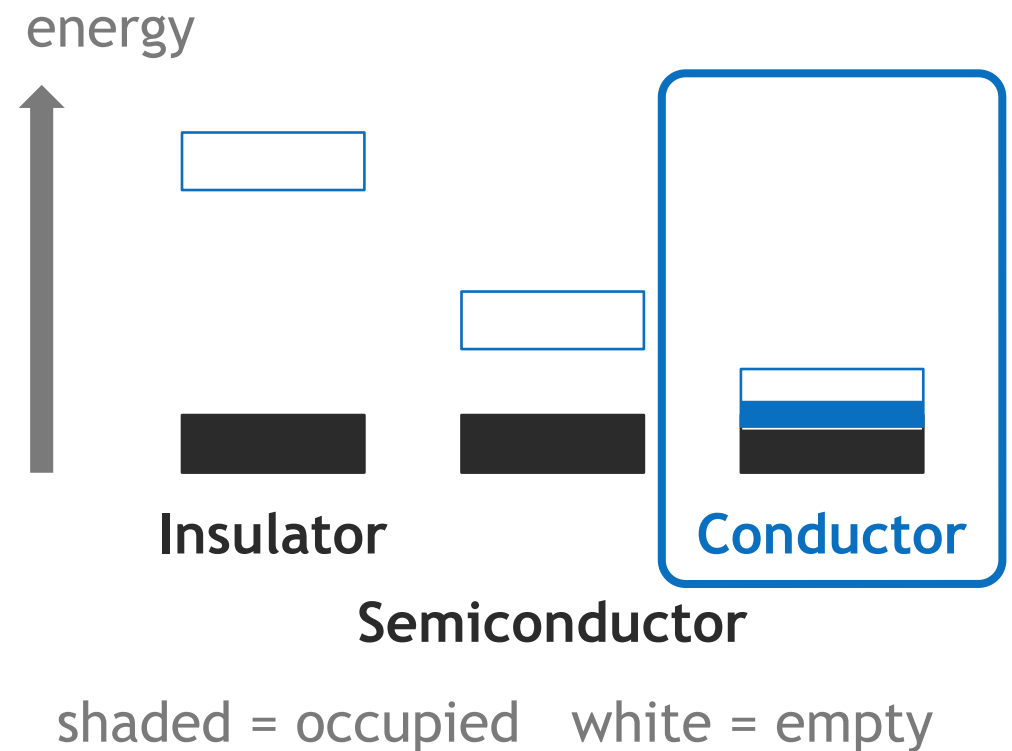




Conductors

In a conductor the conduction band and the valence band overlap, or the highest occupied band is only partly full.

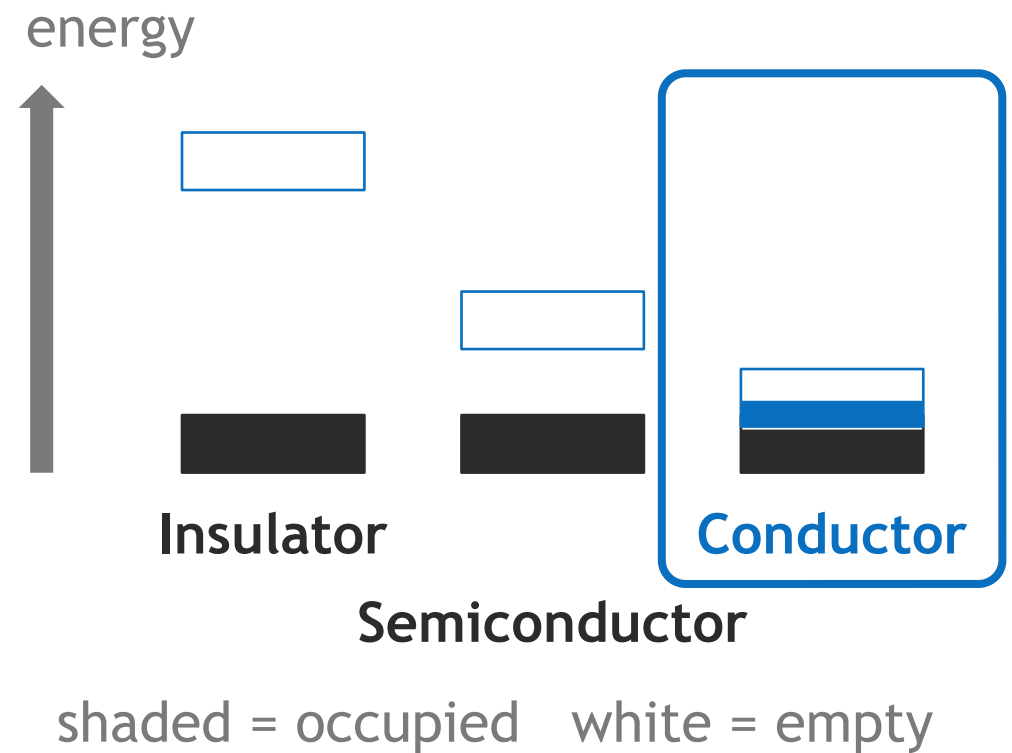
There are electrons free to move AND empty states for them to move into.





Why a conductor conducts well

There is no gap to cross, so a conductor conducts very well.



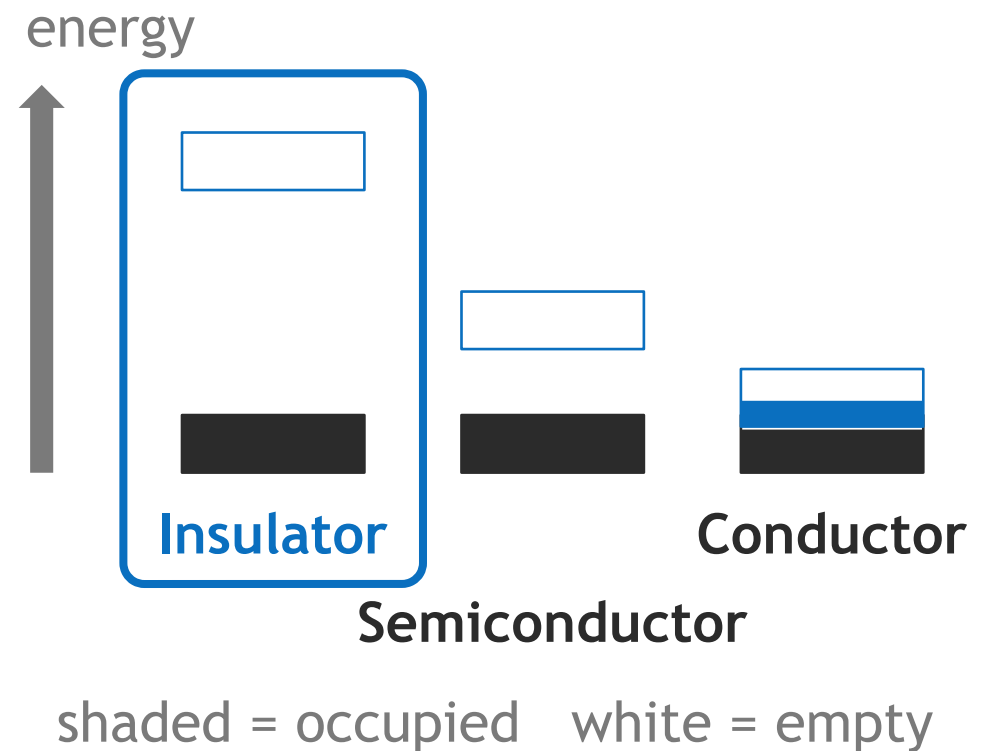


Insulators

In an insulator the valence band is full and the gap above it is very large.

At room temperature no electron has enough energy to cross that gap.

The conduction band stays empty, so no current flows.

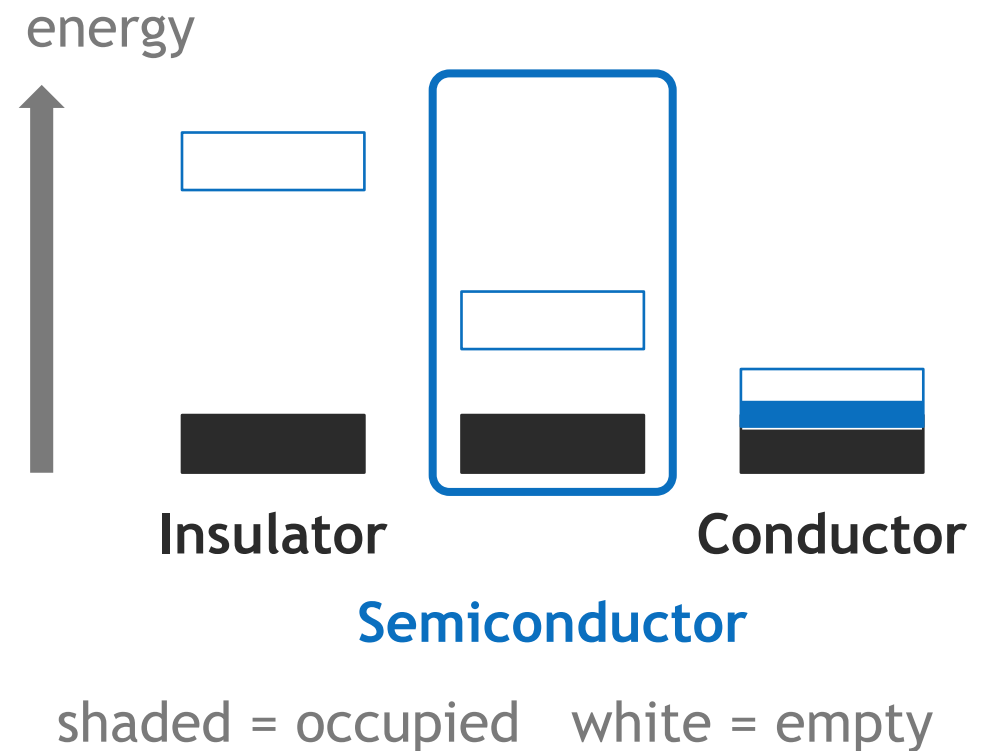




Semiconductors

In a semiconductor the valence band is full, but the gap above it is small.

At room temperature a few electrons have enough energy to cross it.

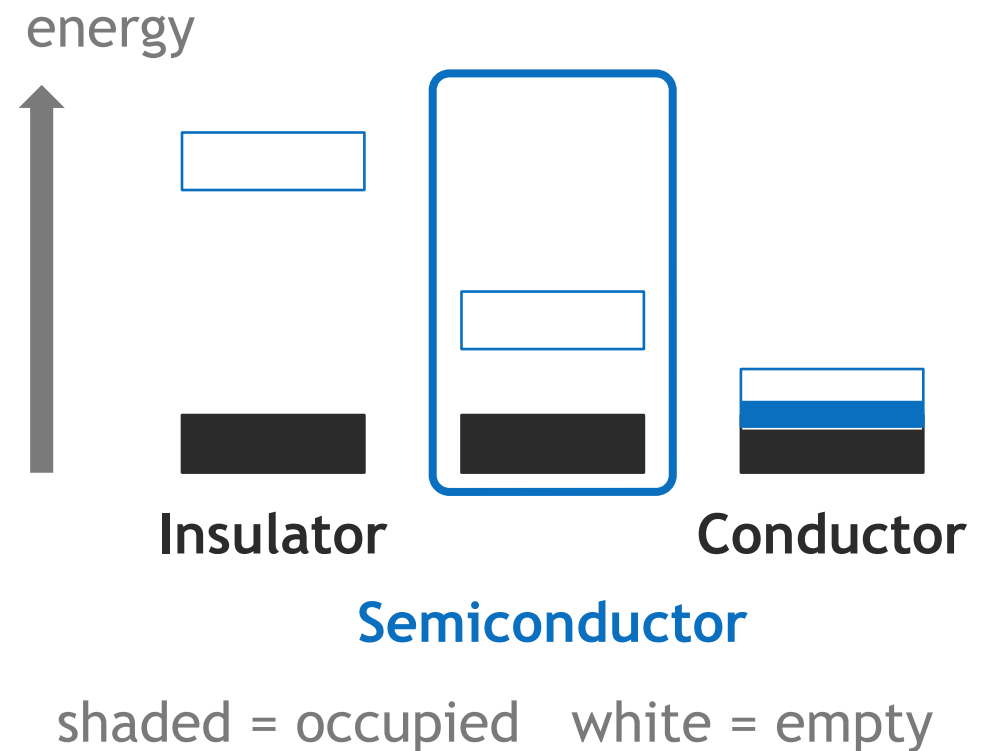




How well a semiconductor conducts

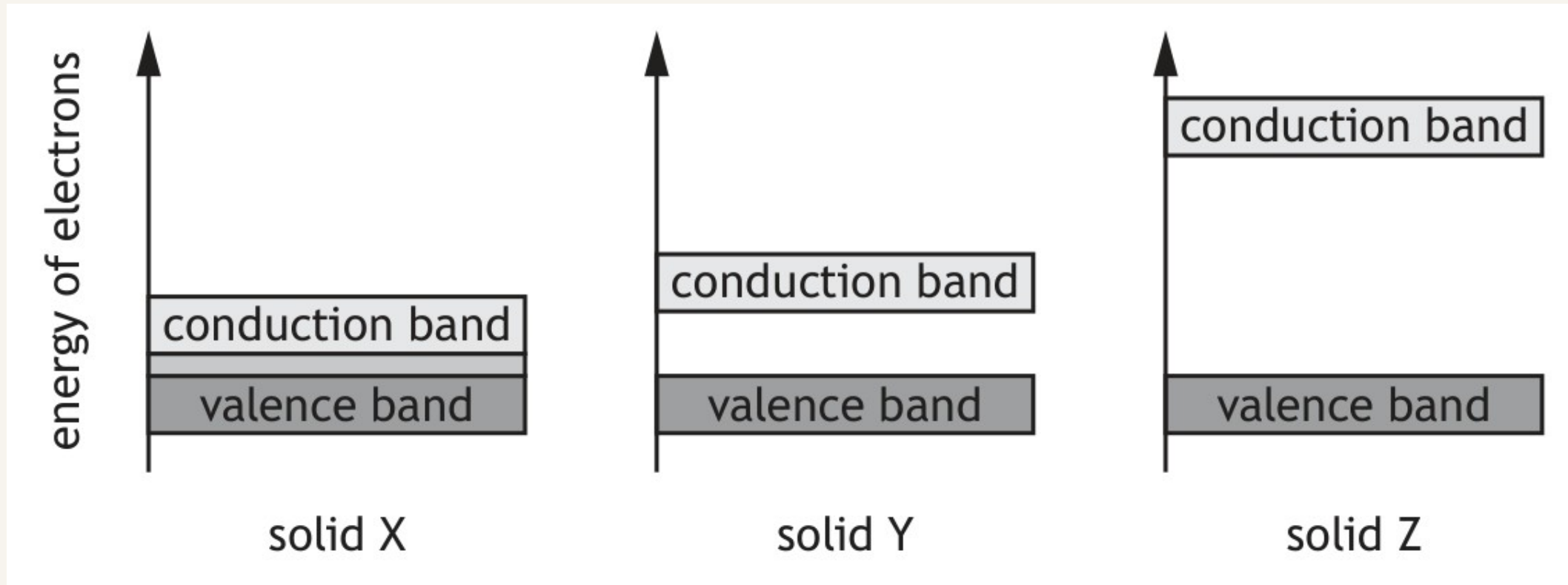
That gives a little conduction — more than an insulator, far less than a metal.

Warm it up and more electrons cross the gap, so a semiconductor conducts better when it is hotter.



The same diagram in an exam

No colour, and the bands are named rather than shaded.



Hinge question: the gap in a semiconductor

SHOW ME BOARDS

In a semiconductor, the energy gap between the valence band and the conduction band is

- A small, allowing some conduction to take place at room temperature
- B large, allowing some conduction to take place at room temperature
- C zero, allowing electrons to move freely
- D large, meaning that no conduction can take place at room temperature
- E small, meaning that no conduction can take place at room temperature.

Show Me: classify the solid

SHOW ME BOARDS

- 1 State which band an electron must reach before it can carry a current.
- 2 A solid has a full valence band and a very large gap. State what kind of solid it is, and why it does not conduct.
- 3 State the two things a solid needs before it can conduct.
- 4 Explain why a semiconductor conducts better when it is warmed up.
- 5 A solid's conduction band and valence band overlap. State what kind of solid it is, and why it conducts.

Task

Work through the band structure questions.
On every explanation, name the bands and
say how big the gap is.

Past papers: 2023 Q23, 2024 Q24



[Past Paper Finder](#)

Starter

2013 Higher Physics

Scan the code and answer these:

Q8

Q11

Section A · charged particles, capacitors

Show your working, not just the letter.



**Scan for the paper
Marking scheme**

© Qualifications Scotland (SQA)

PART 2

Doping

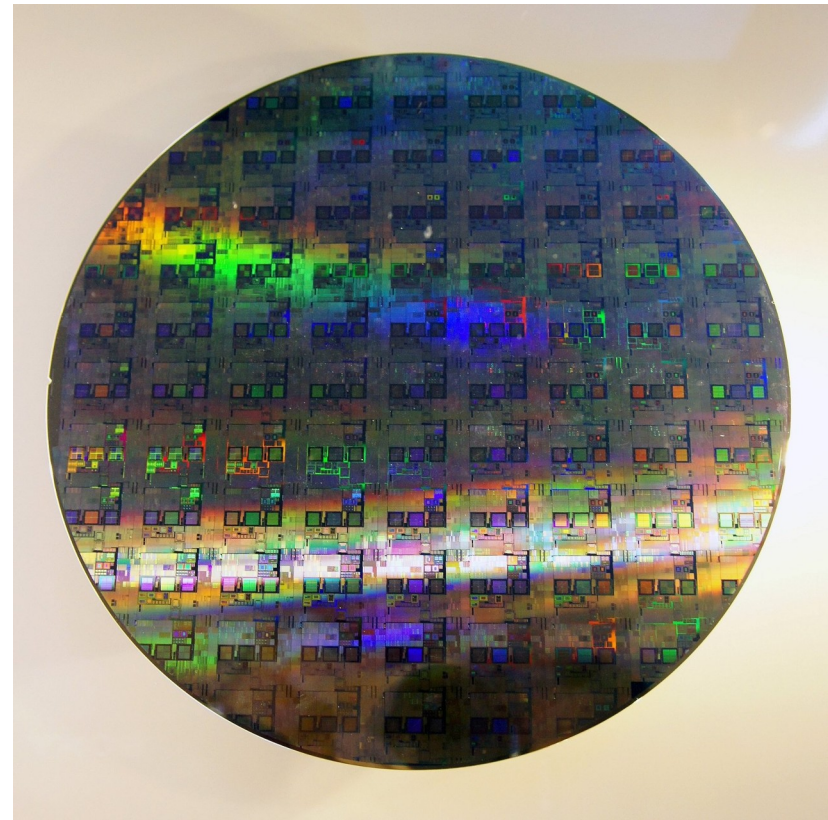
How can adding a tiny amount of impurity make a semiconductor conduct better?

By the end of this part...

- 1 I can describe what doping is and state why it is done.
- 2 I can tell p-type and n-type semiconductors apart.
- 3 I can explain both types using their band structure.

What doping is

- Pure silicon conducts very little at room temperature.
- Chip makers add a tiny amount of impurity.
- Chips start as doped wafers like this one.

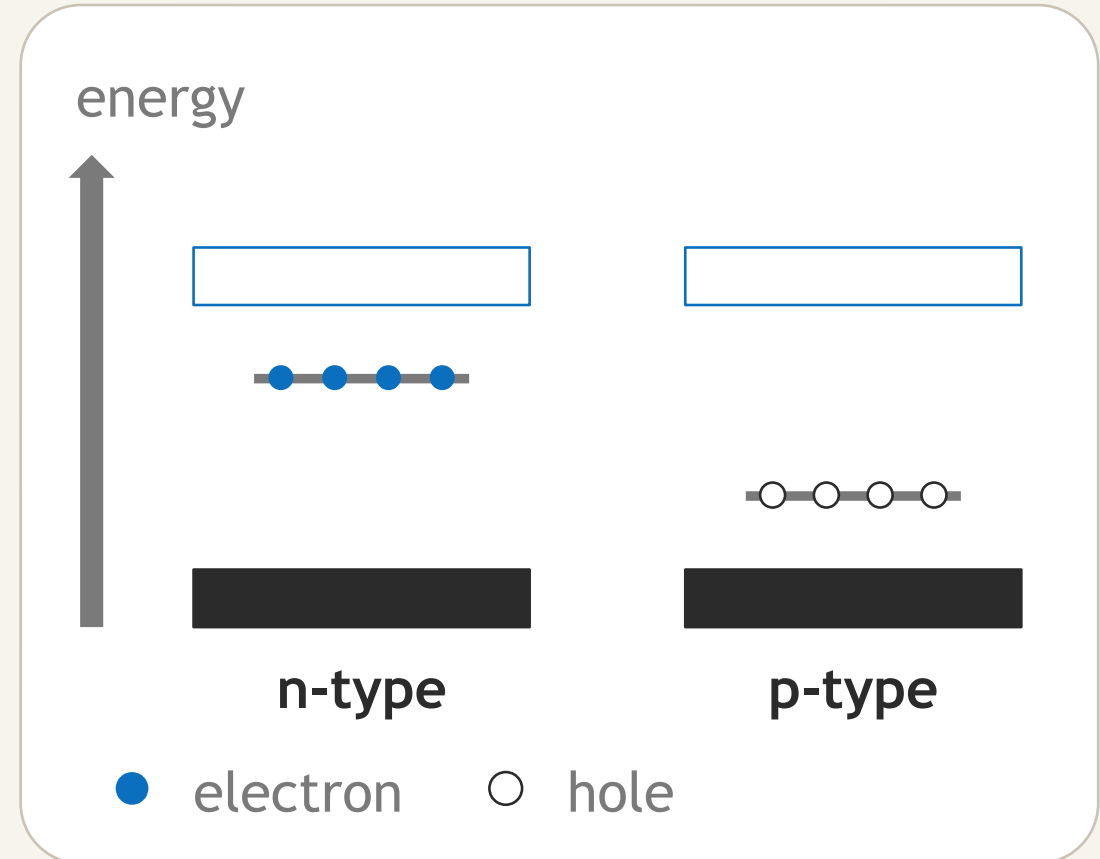




Doping

Doping means adding a small amount of a specific impurity to a semiconductor during manufacture.

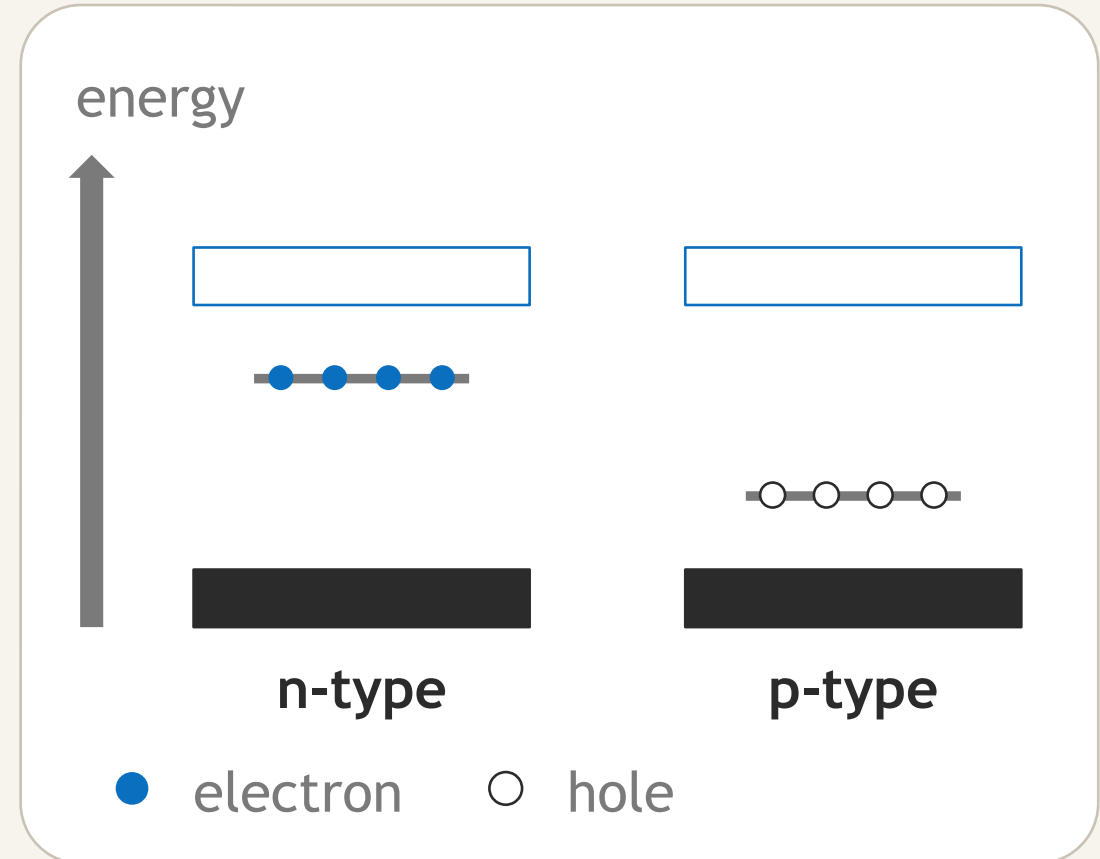
It is done to increase the conductivity in a controlled way.





Two types of doped semiconductor

Doping produces two types of semiconductor: n-type and p-type.

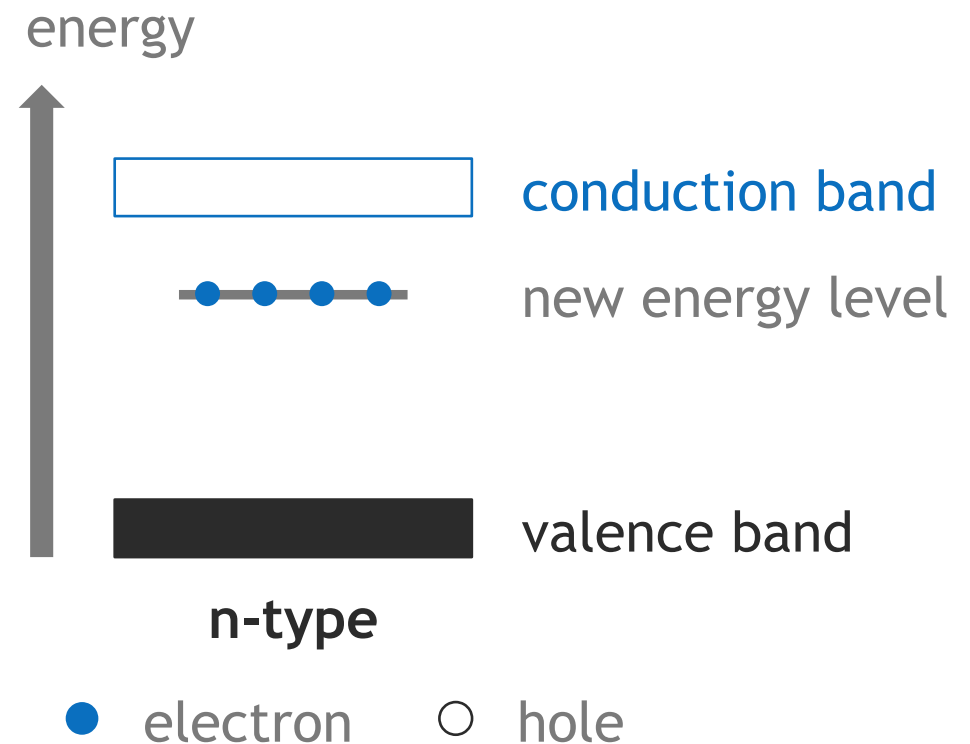




n-type semiconductors

In an n-type semiconductor the new level sits just below the conduction band.

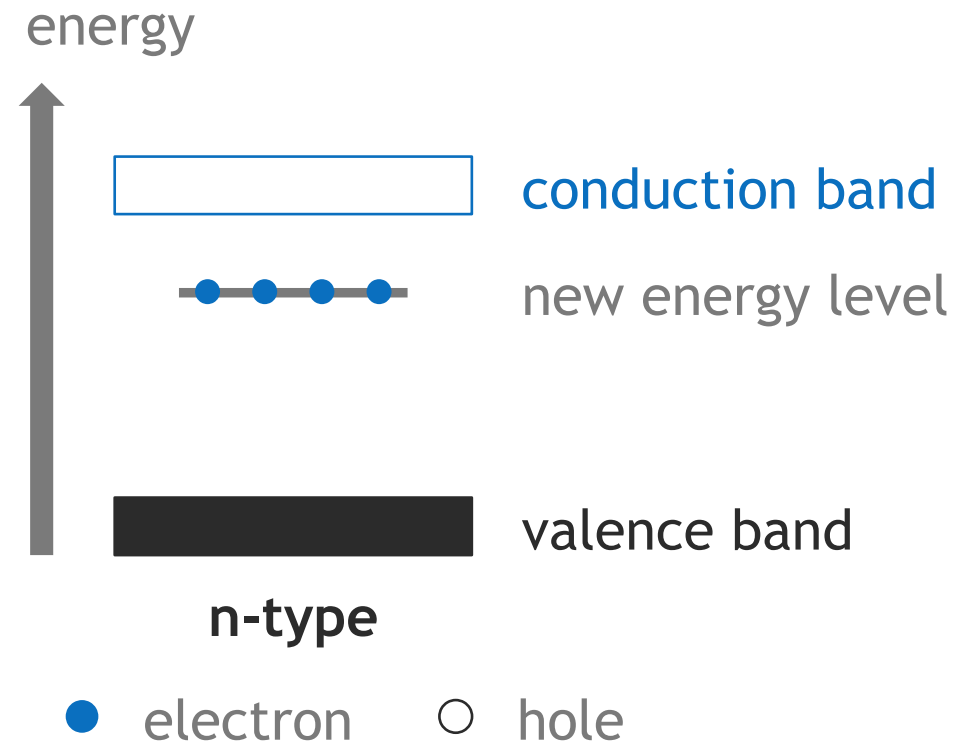
Electrons on that level need only a little energy to reach the conduction band.





Conduction in n-type

So conduction happens by electrons in the conduction band, and the charge carriers are negative.

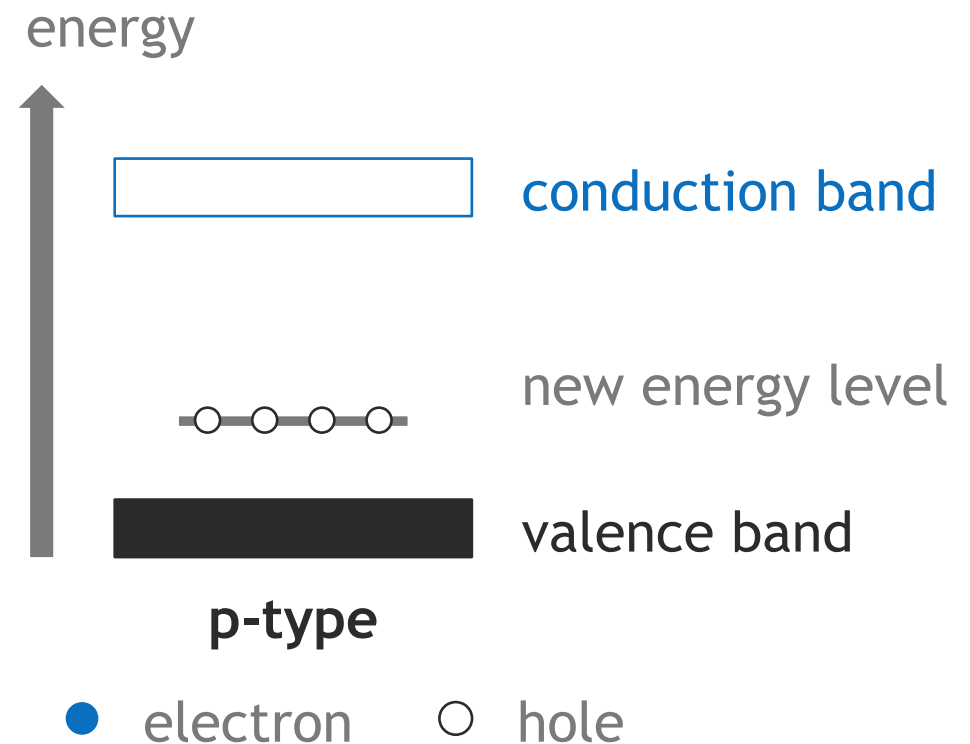




p-type semiconductors

In a p-type semiconductor the new level sits just above the valence band, and it is empty.

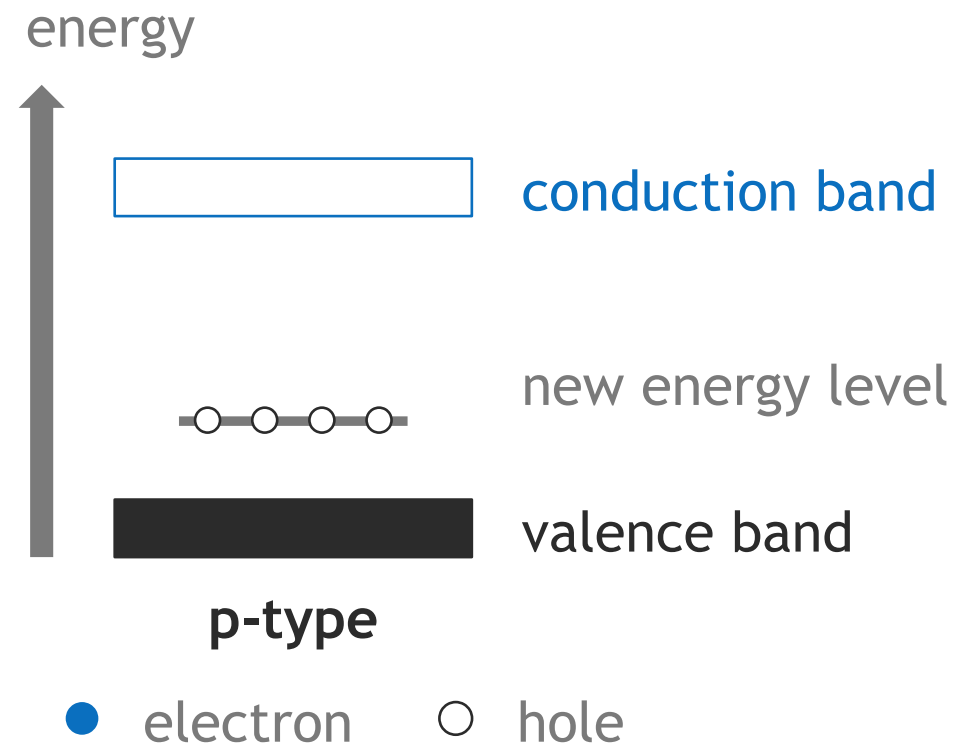
Electrons in the valence band move up into those empty states, leaving holes behind them.





Conduction in p-type

So conduction happens in the valence band, and the charge carriers are holes, which behave as if they are positive.



Hinge question: doping silicon

SHOW ME BOARDS

A crystal of silicon is “doped” with arsenic. This means that a small number of the silicon atoms are replaced with arsenic atoms.

The effect of the doping on the crystal is to

- A make it into a photodiode
- B make it into an insulator
- C increase its resistance
- D decrease its resistance
- E allow it to conduct in only one direction.

Show Me: doping

SHOW ME BOARDS

- 1 State what is meant by doping a semiconductor.
- 2 State why a semiconductor is doped.
- 3 A semiconductor's charge carriers are electrons. State whether it is p-type or n-type.
- 4 State where the new energy level sits in an n-type semiconductor.
- 5 Explain, in terms of bands, how conduction happens in a p-type semiconductor.

Task

Work through the doping questions.
Every answer names a band and says where
the new level sits.

Past papers: 2011 Q18, 2014 Q18



[Past Paper Finder](#)

Starter

2015 Revised Higher Physics

Scan the code and answer these:

Q17

Q19

Q20

Section A · a.c., semiconductors
Show your working, not just the letter.



**Scan for the paper
Marking scheme**

© Qualifications Scotland (SQA)

PART 3

The p-n junction

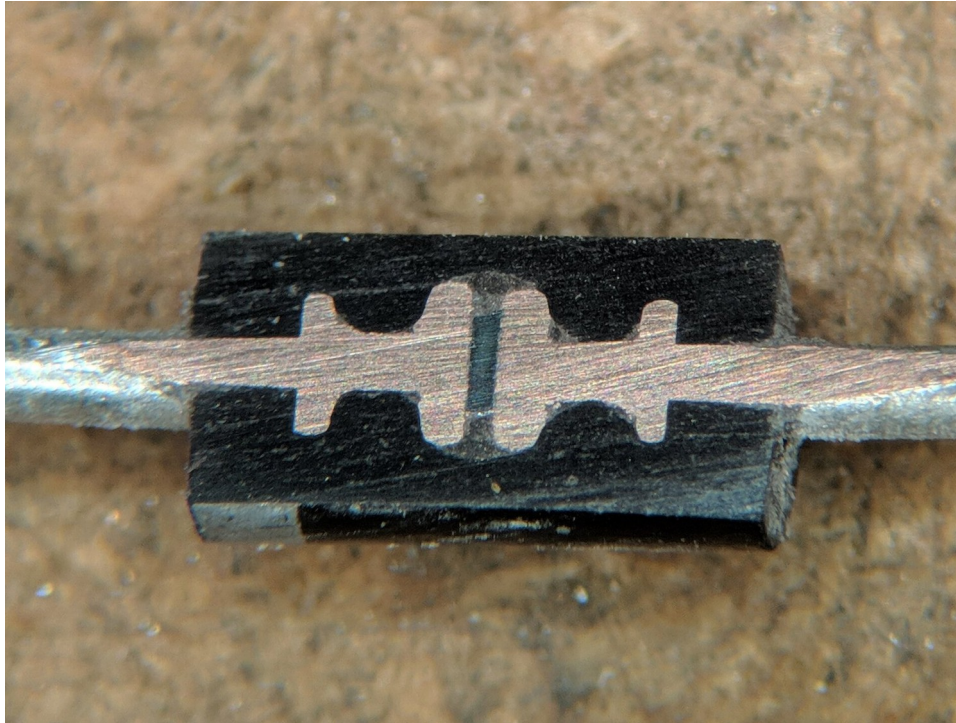
What happens where p-type meets n-type, and why does a diode pass current only one way?

By the end of this part...

- 1 I can describe how a p-n junction is formed and what it contains.
- 2 I can state what forward bias and reverse bias mean.
- 3 I can state the effect of each on the electric field in the junction.

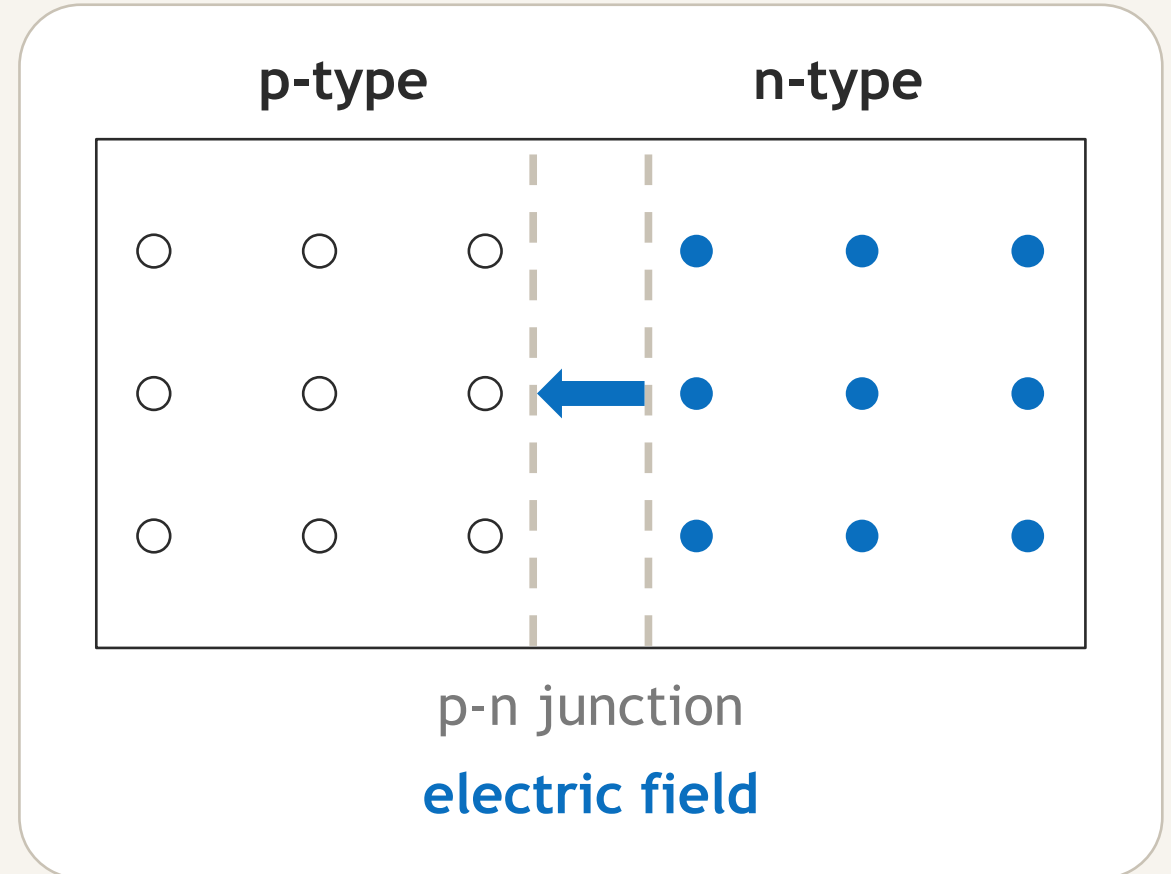
Inside a diode

A diode cut in half. Where is its p-n junction?



How the junction forms

- One piece: p-type at one end, n-type at the other.
- Where the two layers meet is the p-n junction.
- The junction contains an electric field.

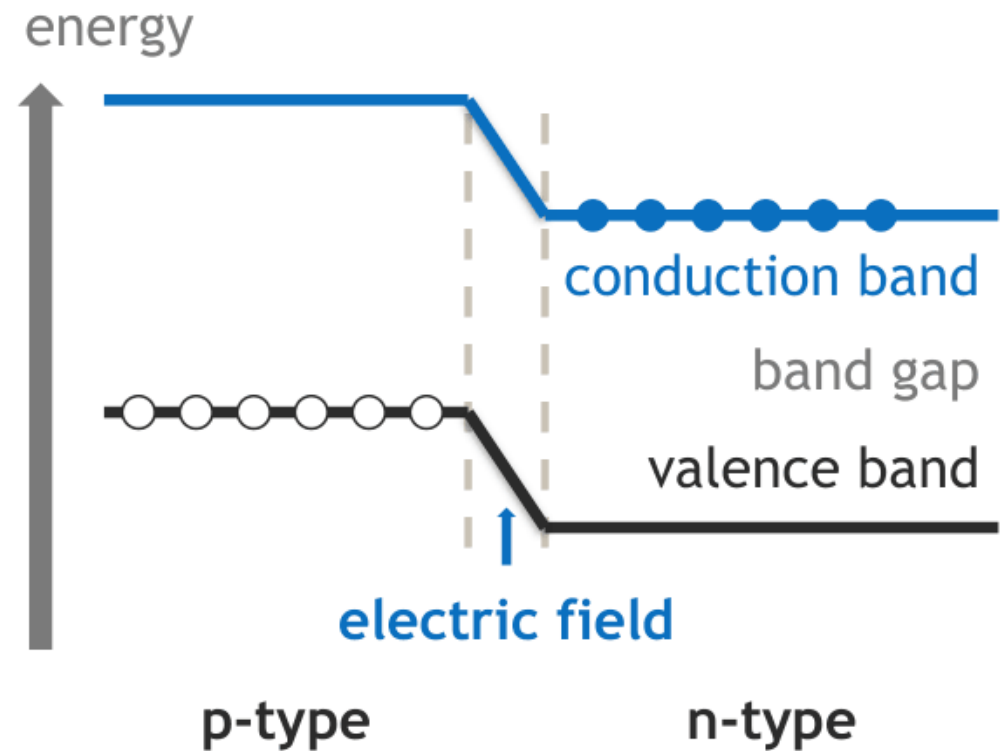




The p-n junction

A p-n junction is formed where a p-type layer and an n-type layer sit next to each other.

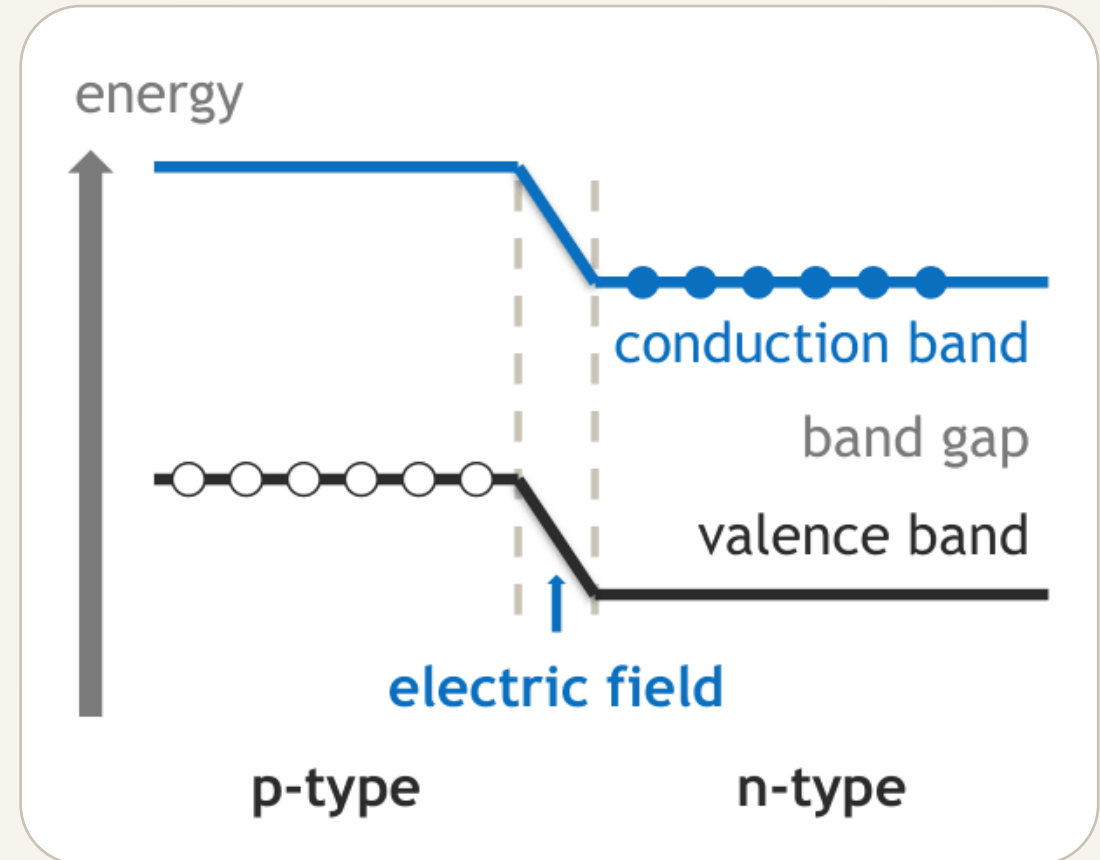
There is an electric field in the p-n junction.





The field shows up as a step

On a band diagram that field shows up as a step: both bands drop as you cross from the p-type side to the n-type side.

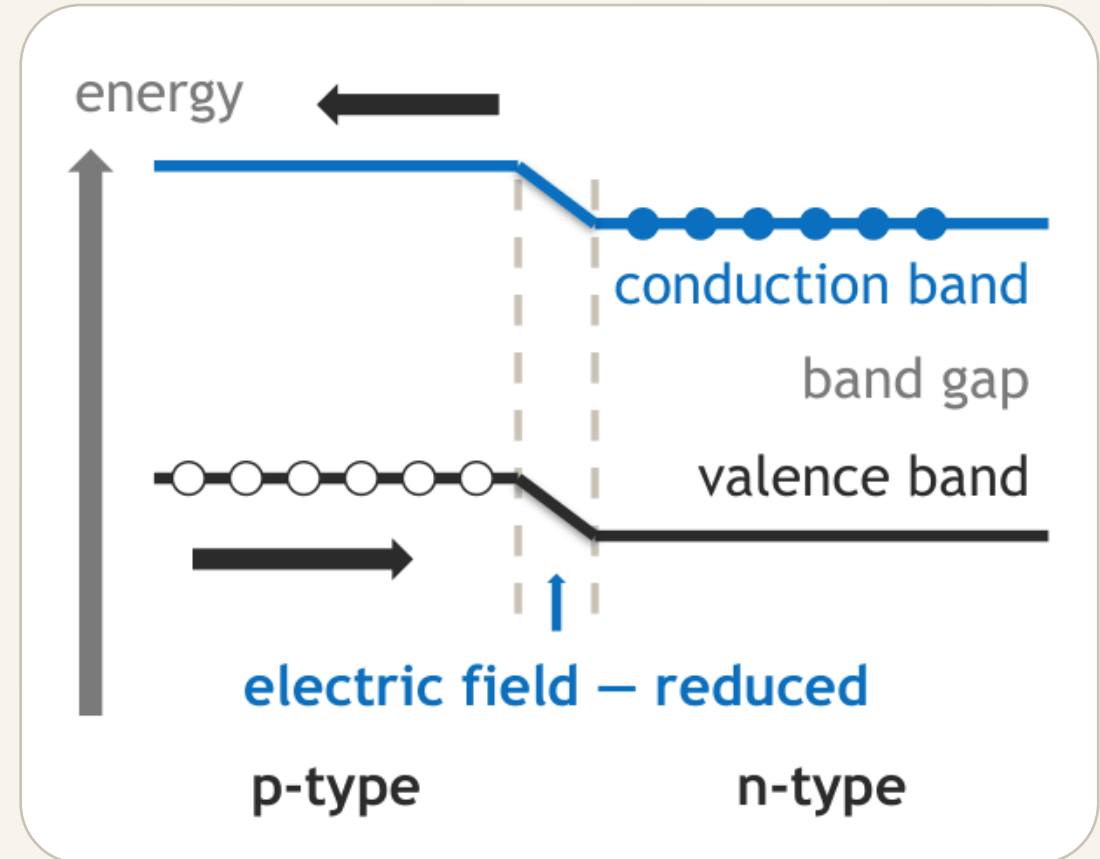




Forward bias

Forward bias connects the supply so that the electric field in the junction is reduced.

On the diagram the step gets smaller, so electrons and holes can cross the junction and a current flows.

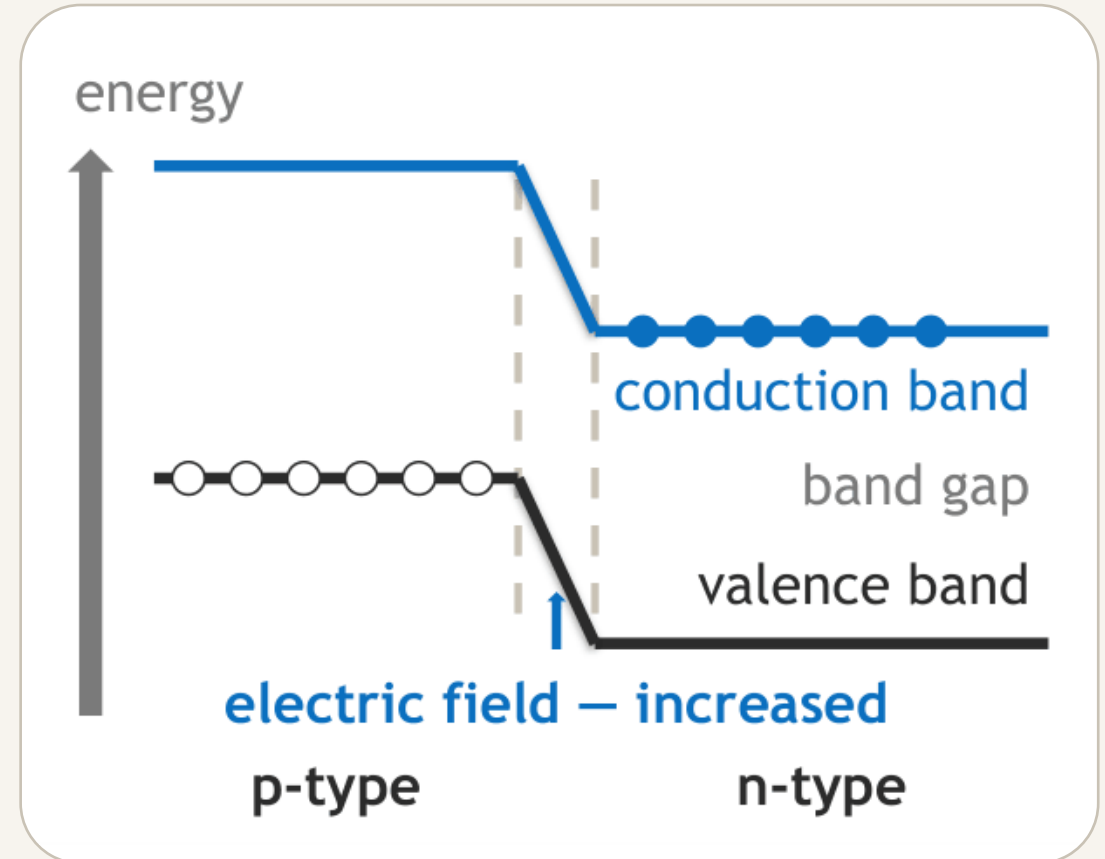




Reverse bias

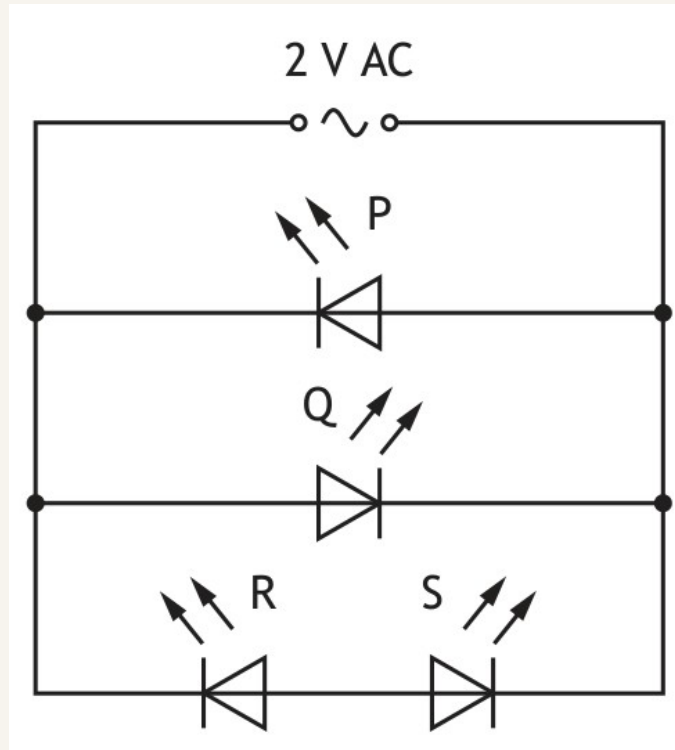
Reverse bias connects it the other way, so the field is increased.

The step gets bigger, nothing can cross, and no current flows.



Bias in a real exam question

Four LEDs on one supply. Which ones are forward biased?

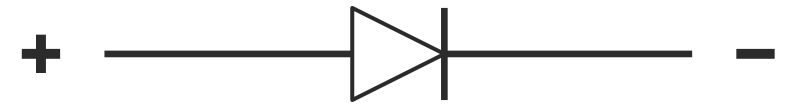




Spotting the bias in a circuit

The triangle in the diode symbol points the way the current is allowed to flow.

If the triangle points from the positive side towards the negative side, the diode is forward biased and it conducts.



forward biased: conducts

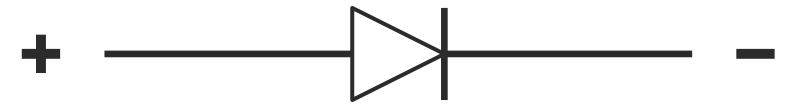


reverse biased: no current



When a diode is reverse biased

Pointing the other way, it is reverse biased and nothing flows.



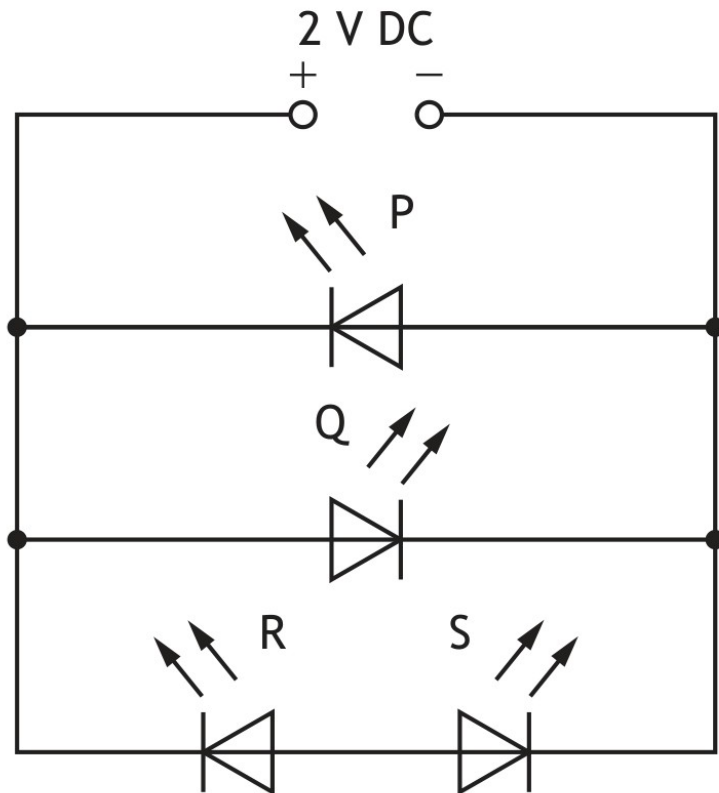
forward biased: conducts



reverse biased: no current

Hinge question: which LEDs light?

SHOW ME BOARDS



A student connects four identical light emitting diodes (LEDs) to a 2 V DC supply as shown.

Which of the LEDs P, Q, R, and S will light?

A P only

B Q only

C P and Q only

D P and R only

E Q and S only

Show Me: junctions and bias

SHOW ME BOARDS

- 1 State what is produced when p-type and n-type layers sit next to each other.
- 2 State what a p-n junction contains.
- 3 State the effect of forward bias on the electric field in a junction.
- 4 State the effect of reverse bias on the electric field in a junction.
- 5 A diode is connected so that no current flows. State how it is biased, and what has happened to the field.

Task

Work through the p-n junction questions.
Sketch the band diagram for each bias from
memory, then check it.

Past papers: 2023 Q5(b)(ii)



[Past Paper Finder](#)

Starter

2014 Higher Physics

Scan the code and answer these:

Q10

Q11

Section A · a.c., capacitors

Show your working, not just the letter.



**Scan for the paper
Marking scheme**

© Qualifications Scotland (SQA)

PART 4

Light-emitting diodes

An LED is a forward-biased p-n junction doing one very specific job: turning electrons into photons.

By the end of this part...

- 1 I can describe an LED as a forward-biased p-n junction diode.
- 2 I can describe, in terms of bands, how an LED emits a photon.
- 3 I can explain why different LEDs emit different colours.

What an LED is

- An LED is a p-n junction diode that gives out light.
- It lights only when it is forward biased.
- Traffic lights, screens and street lamps all use them.

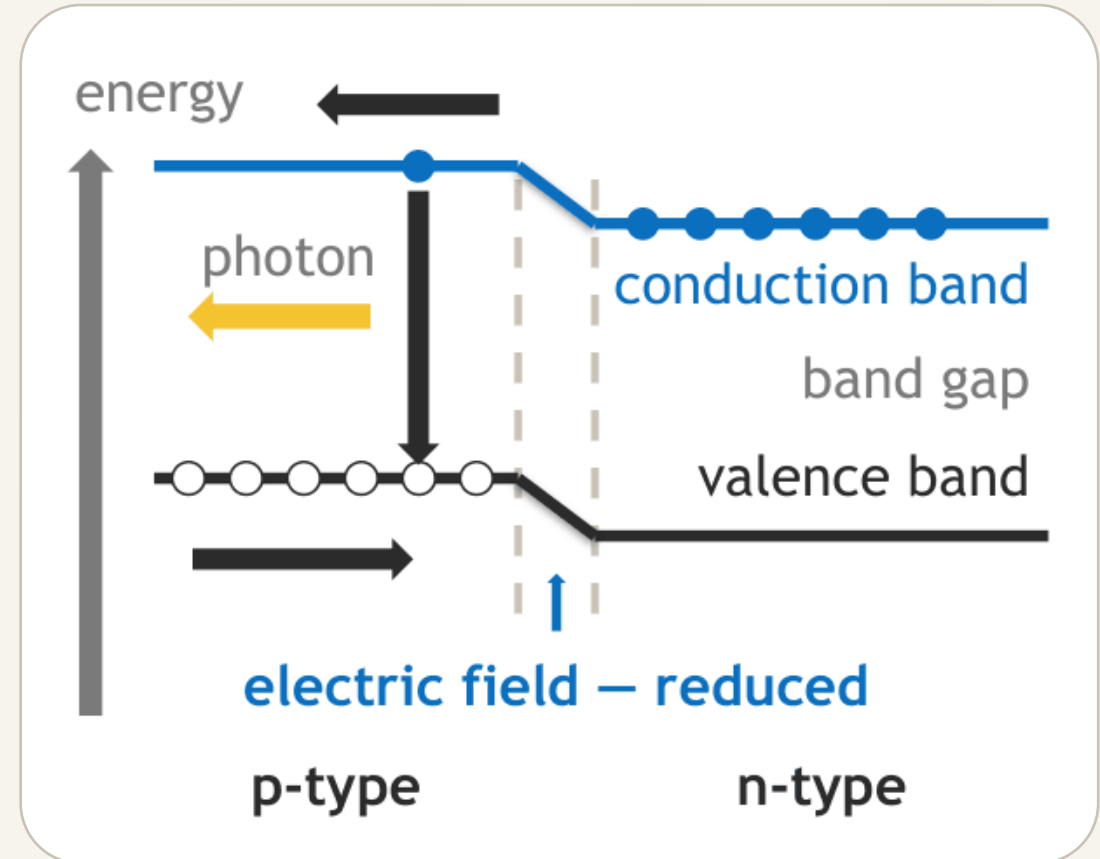




Light-emitting diodes

An LED is a forward-biased p-n junction diode that emits photons.

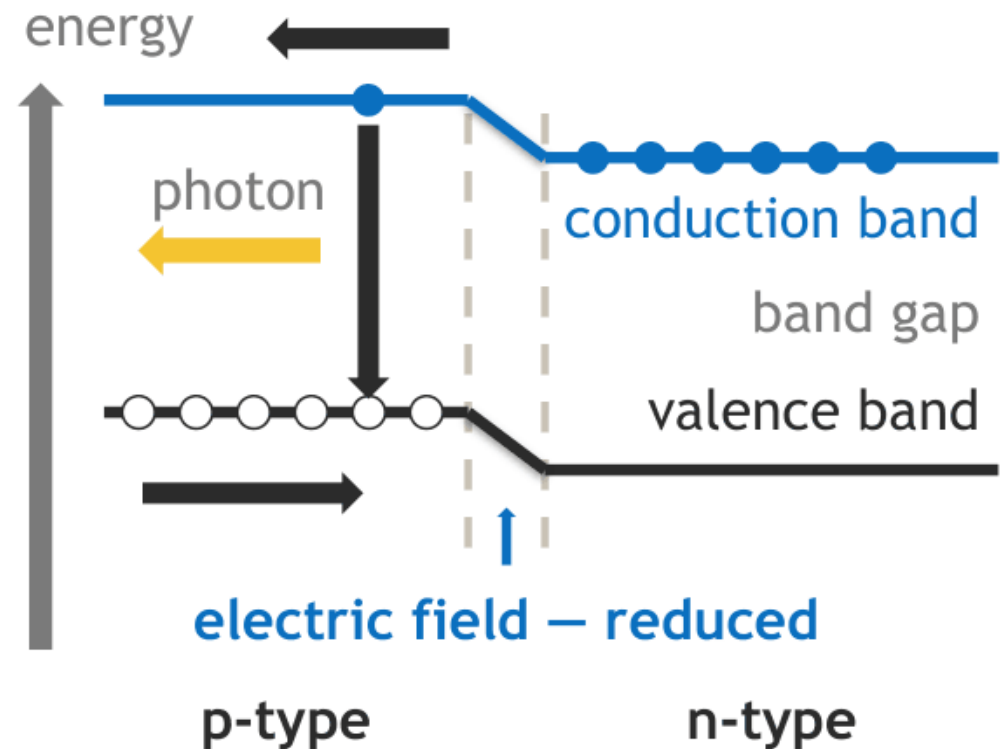
The forward bias moves electrons from the conduction band on the n-type side towards the conduction band on the p-type side.





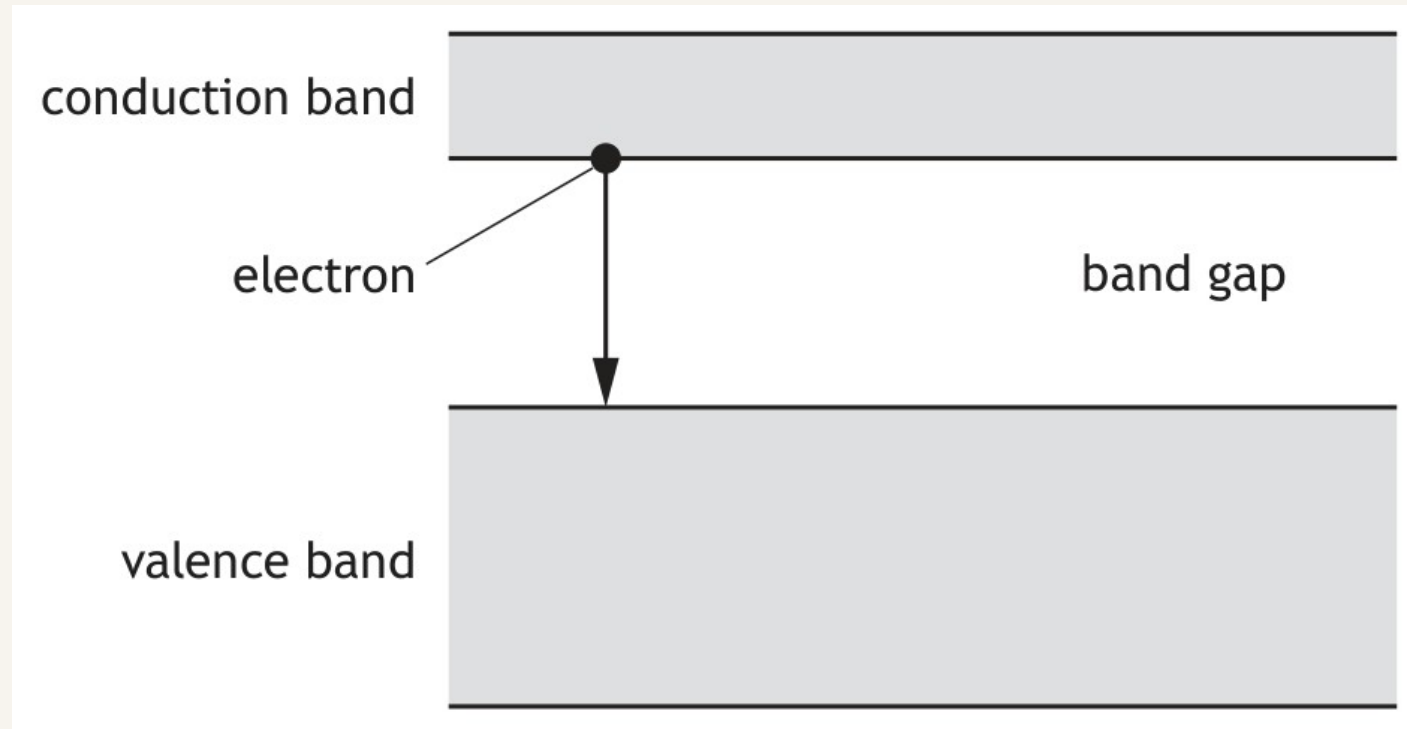
How an LED emits a photon

A photon is emitted when an electron falls from the conduction band into the valence band, either side of the junction.



The same idea in an exam question

One electron, one drop across the gap, one photon.

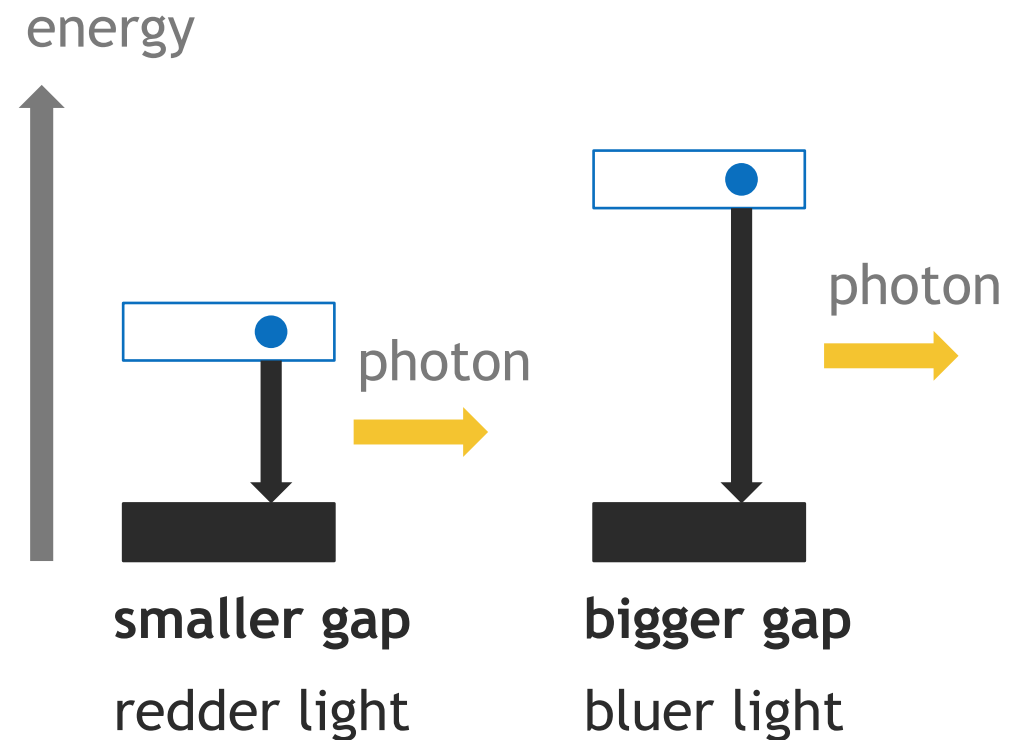




Band gap and colour

The electron loses exactly the energy of the gap as it falls.

All of that energy leaves as one photon, so the photon's energy is equal to the band gap.

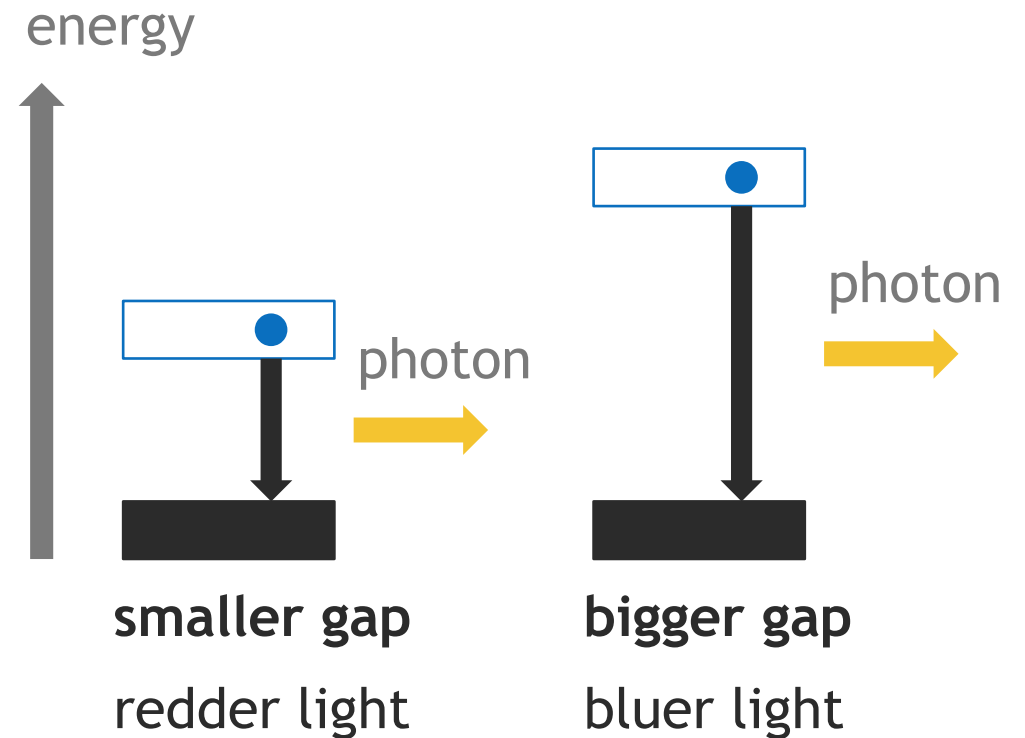




Bigger gap, bluer light

A bigger gap gives a higher-energy photon, which is bluer light.

A smaller gap gives a lower-energy photon, which is redder light.



Hinge question: red and blue LEDs

SHOW ME BOARDS

A student makes the following statements about red and blue LEDs.

- I Red LEDs emit photons of longer wavelength than blue LEDs.
- II The gap between the valence band and the conduction band is larger for red LEDs than blue LEDs.
- III The potential difference across an LED determines the wavelength of the photons emitted.

Which of these statements is/are correct?

A I only

B II only

C III only

D I and II only

E I, II and III

Watch Out – always mention the band gap

THE MISTAKE

- Explaining LED colour without mentioning the band gap
- Treating every LED's photon energy as the same

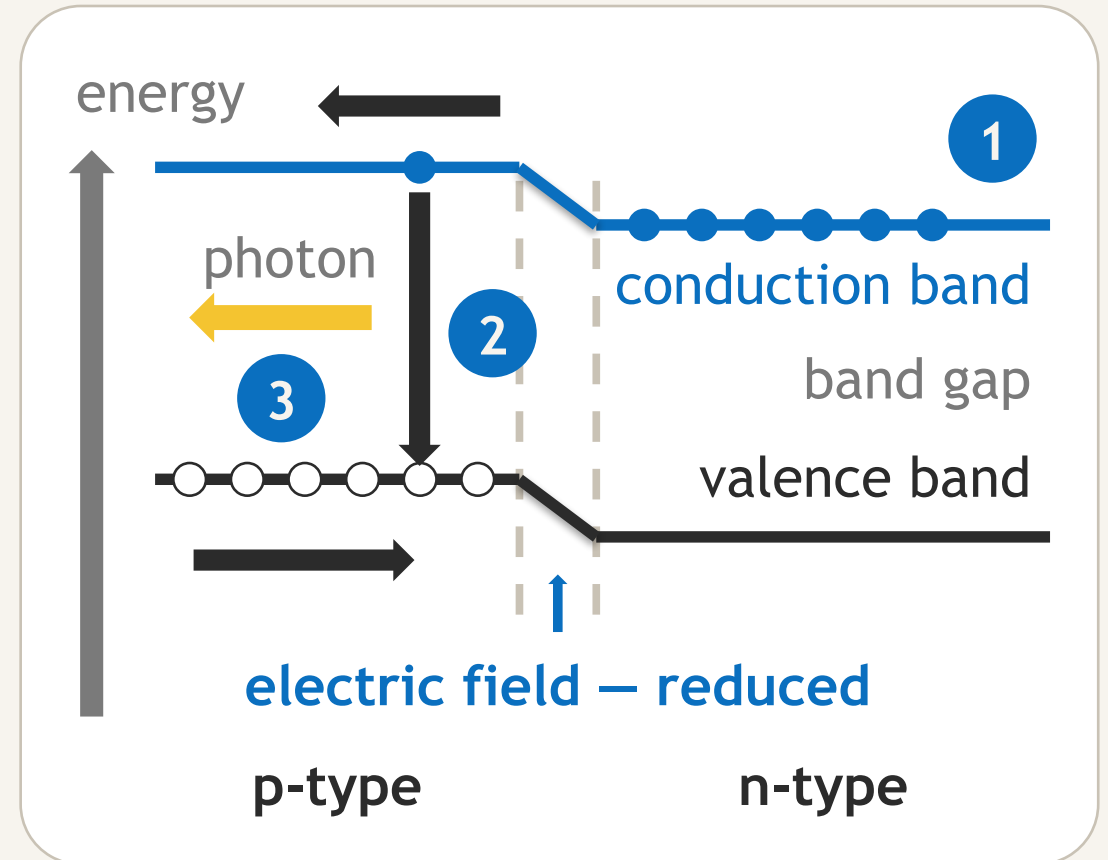
THE CORRECT ANSWER

- Photon energy equals the band gap. Say so.
- A different colour means a different gap.

A 3-mark LED answer: what each mark needs

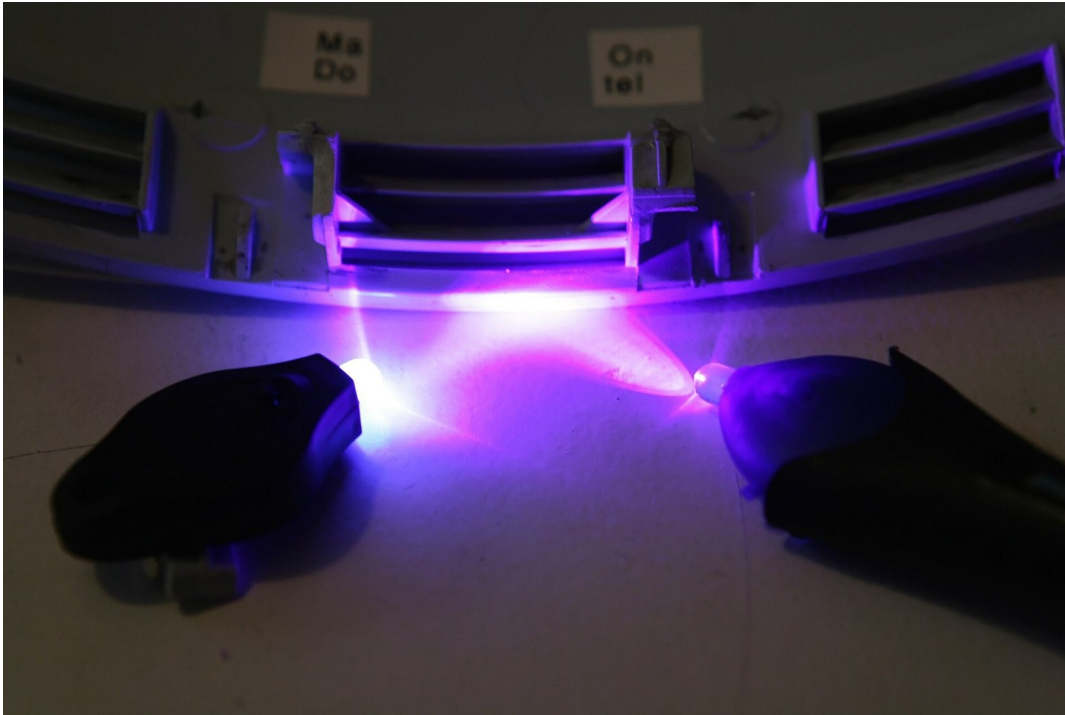
- 1 Electrons move from the n-type towards the p-type conduction band.
- 2 They fall from the conduction band into the valence band.
- 3 A photon is emitted.

Name the bands in full. “Holes and electrons recombine” alone scores 0.



Where you meet LEDs

Which of these LEDs has the bigger band gap? How do you know?



ultraviolet LEDs curing glue



an LED street lamp

Show Me: LEDs

SHOW ME BOARDS

- 1 State what an LED is, in terms of a p-n junction.
- 2 State how an LED must be biased before it will emit light.
- 3 Describe, in terms of bands, how an LED emits a photon.
- 4 Two LEDs emit red and blue light. State and justify which has the larger band gap.
- 5 Explain why one LED emits a single colour rather than a range of colours.

Task

Work through the LED questions.
Every band-theory answer names both
bands and the gap.

Past papers: 2025 Q23, 2026 Q7(c)



[Past Paper Finder](#)

Starter

2014 Revised Higher Physics

Scan the code and answer these:

Q8

Q16

Q18

Section A · particles, semiconductors

Show your working, not just the letter.



**Scan for the paper
Marking scheme**

© Qualifications Scotland (SQA)

PART 5

Solar cells

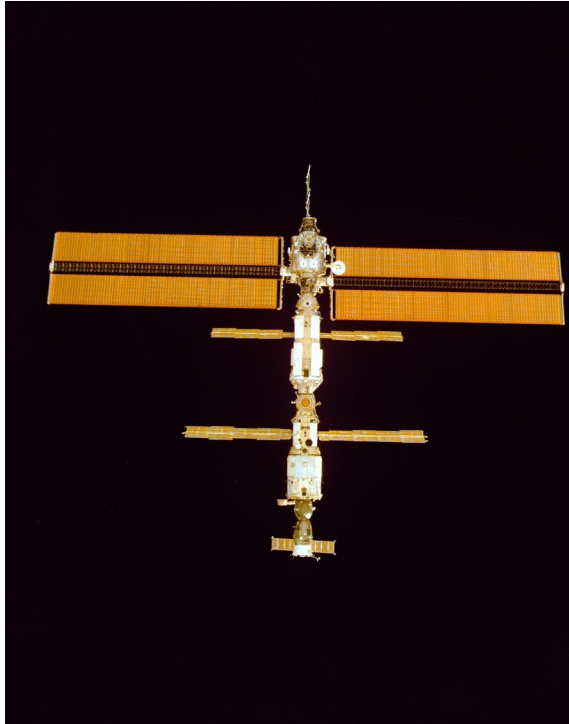
Run the LED backwards — light goes in instead of coming out — and you have a solar cell.

By the end of this part...

- 1 I can describe a solar cell as a p-n junction and name the photovoltaic effect.
- 2 I can describe, in terms of bands, how a solar cell produces a p.d.
- 3 I can state the difference between a solar cell and an LED.

Powered by light

The Space Station runs on solar cells. What happens inside one?



The photovoltaic effect

- A solar cell turns light into electrical energy.
- It is a p-n junction, just like an LED.
- Panels join many cells, like these in Midlothian.

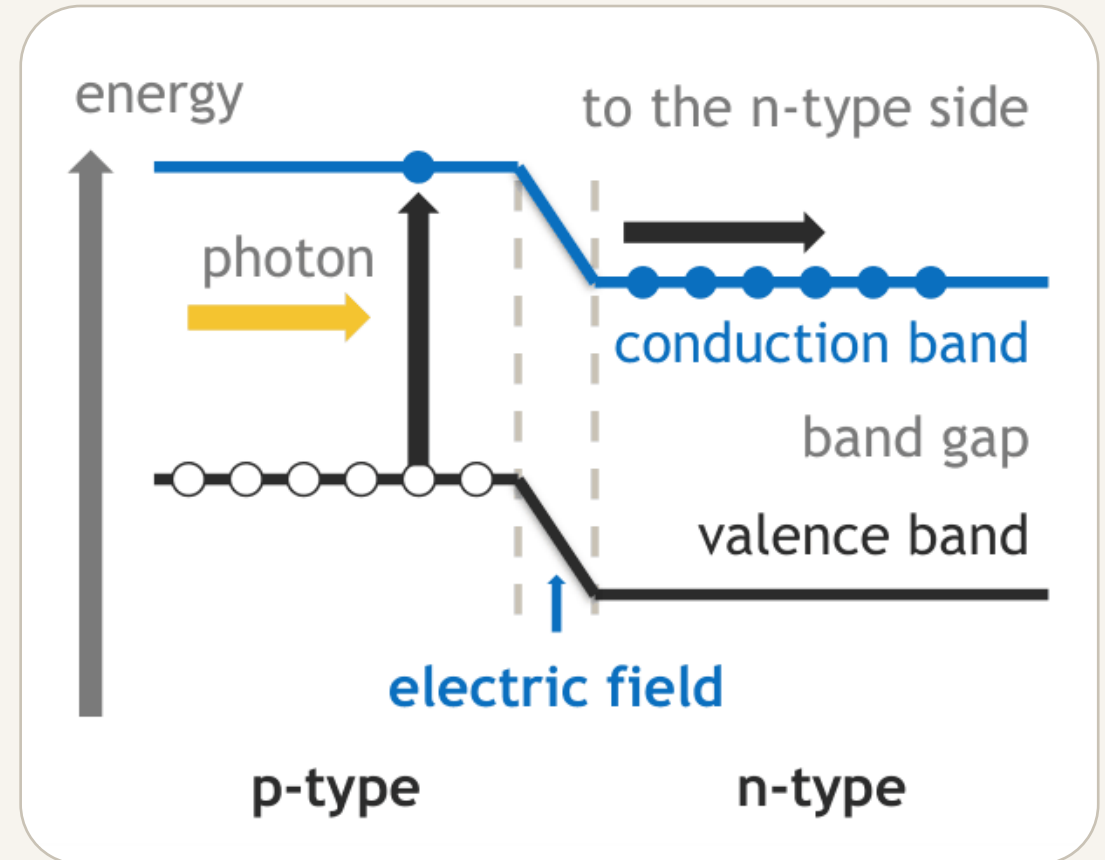




Solar cells

A solar cell is a p-n junction designed to produce a potential difference when photons are absorbed.

This is called the photovoltaic effect.

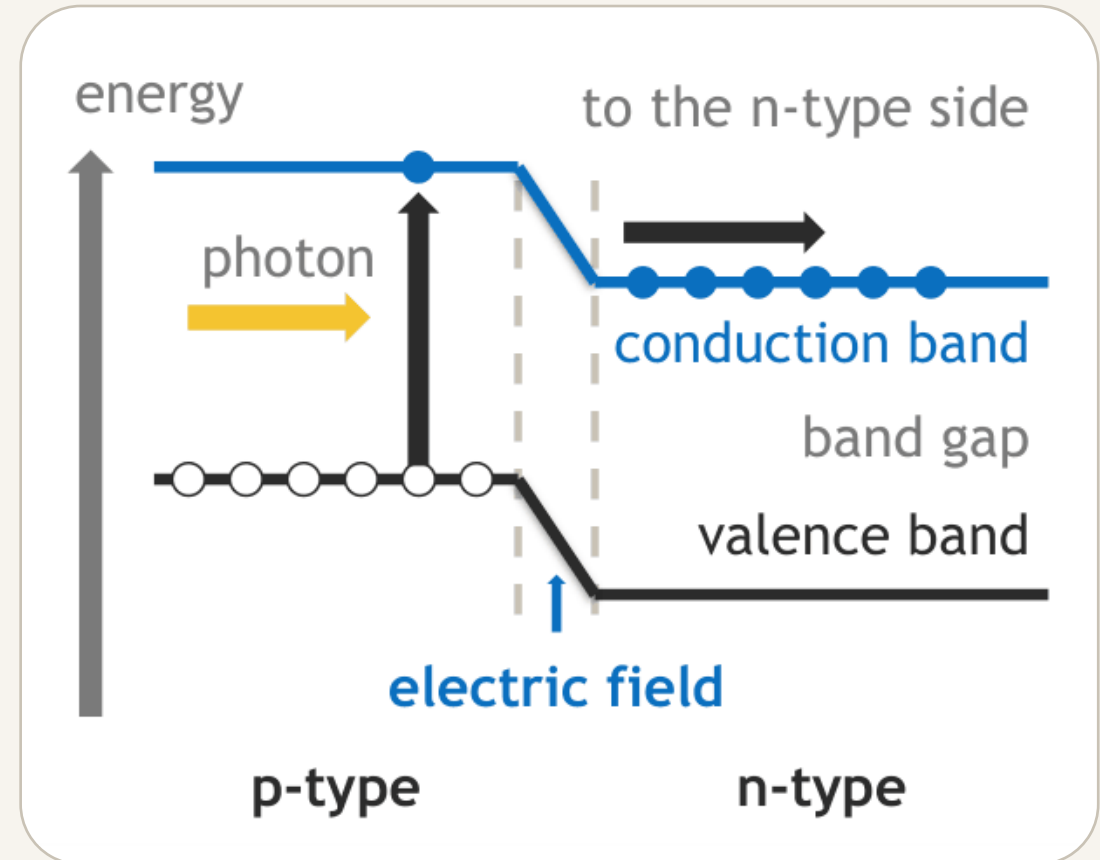




From photon to moving electron

Absorbing a photon gives an electron enough energy to be raised from the valence band to the conduction band.

Once the electron is in the conduction band, the junction's electric field moves it towards the n-type side.

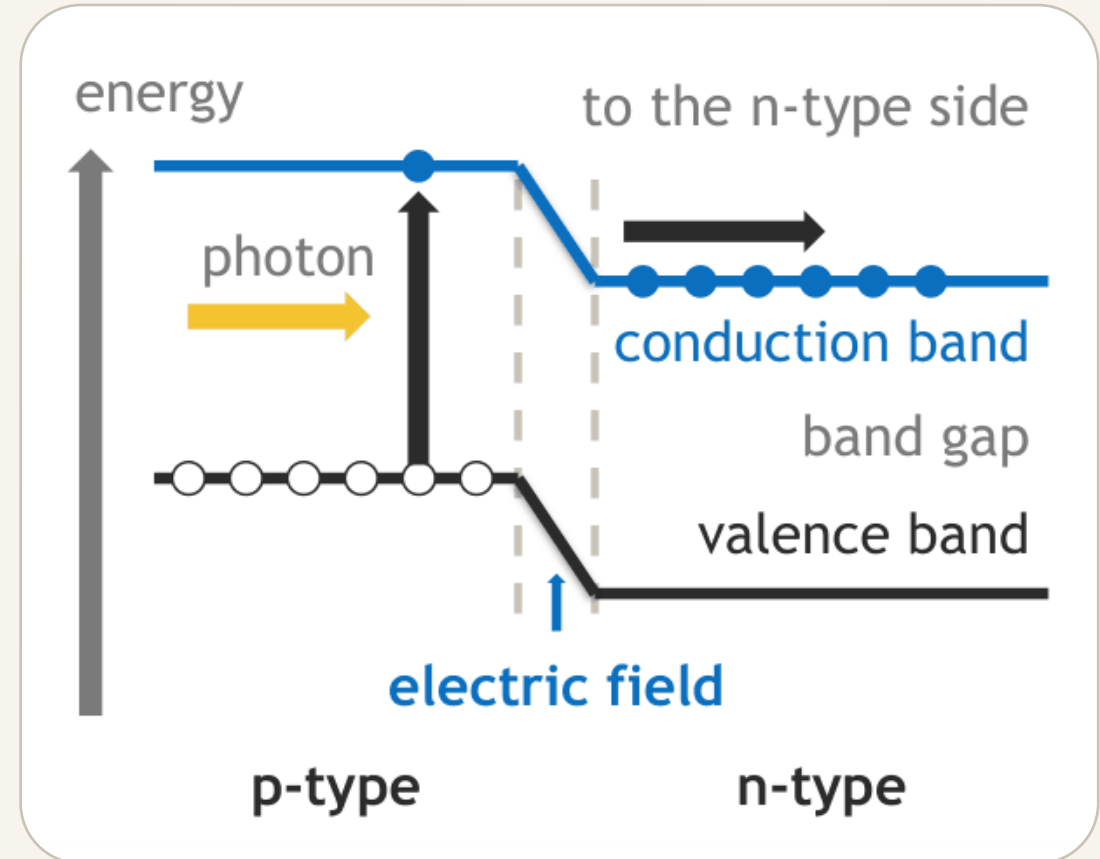




How the potential difference appears

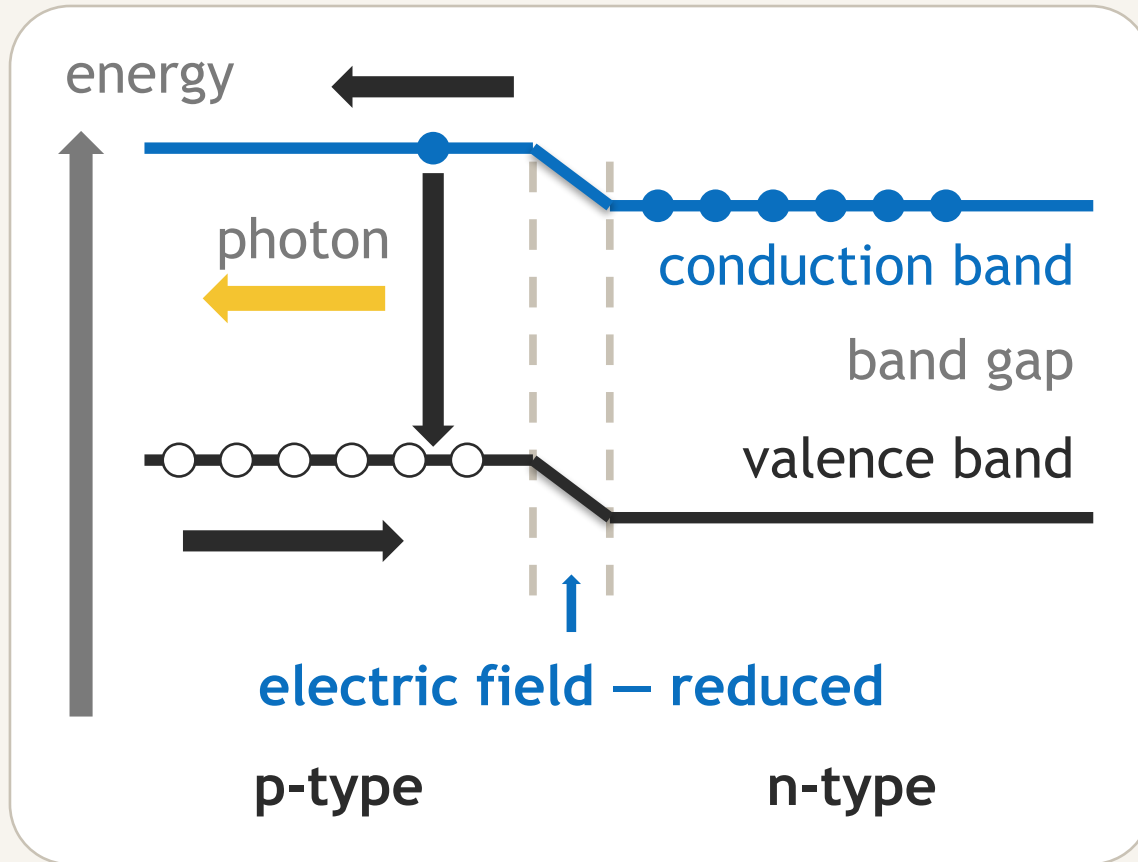
Electrons gathering on one side and not the other is what produces a potential difference.

That potential difference appears across the whole solar cell.

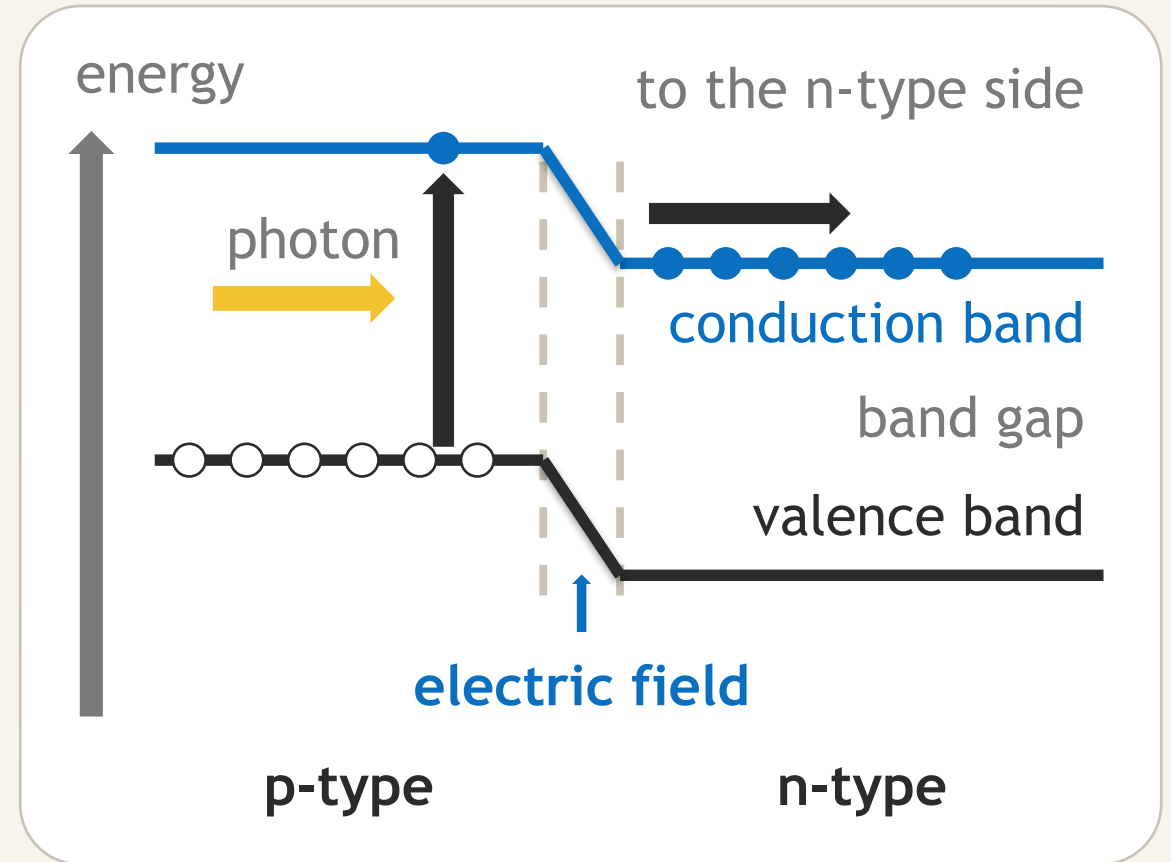


LED or solar cell?

LED: photon out



Solar cell: photon in



Watch Out — solar cell, not LED

THE MISTAKE

- Describing a solar cell as if it emits photons
- Mixing up which device absorbs and which emits

THE CORRECT ANSWER

- A solar cell ABSORBS a photon; an LED EMITS one.
- Check which device the question actually names.

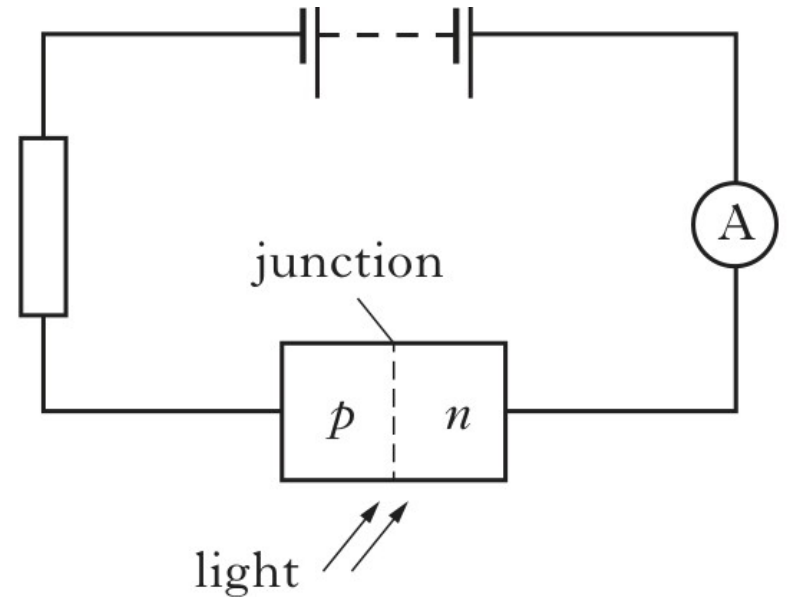


Reading the circuit version

The box marked p and n is the junction itself, with the dashed line showing where the two layers meet.

The arrows show light falling on the junction.

Light in gives a potential difference out, which is what the meter in the circuit reads.



Hinge question: p-n junction devices

SHOW ME BOARDS

A student makes the following statements about p-n junction devices.

- I In solar cells, a potential difference is produced when photons are incident on the junction.
- II The photovoltaic effect occurs in solar cells.
- III In LEDs, photons are emitted from the junction when a current is passed through it.

Which of these statements is/are correct?

- A I only
- B III only
- C I and II only
- D I and III only
- E I, II and III

Show Me: solar cells

SHOW ME BOARDS

- 1 Name the effect by which a solar cell produces a potential difference.
- 2 State what happens to an electron when a solar cell absorbs a photon.
- 3 Describe, in terms of bands and the junction's field, how a solar cell produces a p.d.
- 4 A solar cell and an LED are both p-n junctions. State the key difference in what each one does.
- 5 Explain why a solar cell produces no potential difference in the dark.

Task

Work through the solar cell questions.
Read the question twice – solar cells and LEDs are easy to swap by mistake.

Past papers: 2017 Q14(b), 2024 Q15(b)



[Past Paper Finder](#)

Experiments you could try

- Put an LED and a resistor across a cell, both ways round: it lights only when forward biased.
- Shine a torch on a red LED connected to a voltmeter: it gives a small p.d.
- Connect a solar cell to a voltmeter under a lamp, then cover it.
- Warm a thermistor in your hand and watch its resistance fall.

Checkpoint

- 1 State the difference between the valence band and the conduction band.
- 2 A solid's highest occupied band is only partly full. State what kind of solid it is, and why it conducts.
- 3 Explain, in terms of band structure, why a semiconductor conducts better when it is warmer.
- 4 State what doping is, and where it puts a new energy level.
- 5 Describe, in terms of bands, how an LED emits a photon.

Exam Practice

2024 Q15(b) — band theory for a solar cell,
not an LED.

2018 Q11(c) — comparing a red LED and a
blue LED.

2026 Q7(c) — doping and the LED.

Past papers, Paper 2

LOOKING AHEAD

Next: Motion, Graphs & Projectiles

Scalars, vectors, and the equations that connect them — the largest topic of the course.

- 1 Name the two bands, and say what the gap between them means.
- 2 Explain, using band structure, why an insulator does not conduct.
- 3 State what doping is, and why a semiconductor is doped.
- 4 State the effect of forward bias on the field in a p-n junction.
- 5 An LED emits light and a solar cell absorbs it. Explain the difference in terms of the electron's journey between the bands.

Extension

SHOW ME BOARDS

- 1 A pure semiconductor is doped n-type. Sketch its band diagram before and after, and explain what changed.
- 2 Solar panels lose efficiency as they get hot, even though semiconductor conductivity rises with temperature. Suggest why.
- 3 Explain, using band diagrams, why a p-n junction passes current one way but not the other.

Question 2: InSight's dusty solar panel on Mars

